Spherical Equations (imported from Spherical.mx)

This document lists the field equations exported by the Spherical computation pipeline. All equations are stored in a single Wolfram Language file, Spherical.mx, as a nested Association. No packages (such as xAct) are required to load or manipulate them.

Importing the equations

To import the equations into a fresh Wolfram Language session, evaluate:

```
SphericalEquations = Import["Spherical.mx"]
```

This returns an Association whose top-level keys name each equation set:

```
Keys[SphericalEquations]
```

Each equation set is itself an Association keyed by coordinate component lists. For example, to access the (τ,τ)component of the Euclidean metric equation of motion:

```
SphericalEquations["EuclideanMetricEOM"][{"CTau", "CTau"}]
```

Scalar quantities (such as the kinetic scalar or the scalar field equation) use an empty list {} as their key:

```
SphericalEquations["EuclideanScalarEOM"][{}]
```

The kinetic scalar X_E is stored directly (not as a sub-association):

```
SphericalEquations["KineticXEScalar"]
```

Each expression returned is an ordinary Wolfram Language expression involving the coordinate scalars CTau[], CRho[], CTheta[], CPhi[] and the functions ArealRadius, MetricA, ClockProfile, FuncG4, together with their derivatives. These can be manipulated with standard tools such as Simplify, Solve, DSolve, etc.

Equation listing

EuclideanMetricEOM

```
SphericalEquations["EuclideanMetricEOM"][{"CTau", "CTau"}]
```

(-16 (4 CouplingC1 + CouplingC2) (-1 + MetricA[CTau[], CRho[]])^3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^4 -
24 (4 CouplingC1 + CouplingC2) ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^3 MetricA$^{(0,1)}$[CTau[], CRho[]] +
16 (8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3) (-1 + MetricA[CTau[], CRho[]])^5
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2) +

8 (4 CouplingC1 + CouplingC2) ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^{4}
(ArealRadius$^{(0,2)}$[CTau[], CRho[]] (2 - 6 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}) -
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{3} MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]]) -
4 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2}
(-4 (4 CouplingC1 + CouplingC2 + 2 (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}) - 2 (4 CouplingC1 + CouplingC2)
ArealRadius[CTau[], CRho[]] (6 ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]) + 4 MetricA[CTau[], CRho[]]^{3}
(4 CouplingC1 + CouplingC2 + 2 (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} - ArealRadius[CTau[], CRho[]]^{2}
FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} +
(4 CouplingC1 + CouplingC2) ArealRadius[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]]) + ArealRadius[CTau[], CRho[]]^{2}
(4 (4 CouplingC1 + CouplingC2) MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} + 8 CouplingC1
MetricA$^{(0,2)}$[CTau[], CRho[]] + 2 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]] +
8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} + 4 FuncG4′[KineticXEScalar[]]
(2 ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} + ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}) -
2 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} - 5 CouplingC2
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} - 12 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
4 CouplingC1 MetricA$^{(2,0)}$[CTau[], CRho[]] - 2 CouplingC2
MetricA$^{(2,0)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA$^{(2,0)}$[CTau[], CRho[]]) +
2 MetricA[CTau[], CRho[]]^{2} (-6 (4 CouplingC1 + CouplingC2 + 2 (2 CouplingC1 +
CouplingC2 + 2 CouplingC3) ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}) -
3 (4 CouplingC1 + CouplingC2) ArealRadius[CTau[], CRho[]]
(2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] - ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^{2} (4 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} +
FuncG4′[KineticXEScalar[]] (4 ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} +
6 ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}) +
(2 CouplingC1 - CouplingC2 - 4 CouplingC3) MetricA$^{(2,0)}$[CTau[], CRho[]])) +
MetricA[CTau[], CRho[]] (12 (4 CouplingC1 + CouplingC2 + 2 (2 CouplingC1 +
CouplingC2 + 2 CouplingC3) ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}) +
12 (4 CouplingC1 + CouplingC2) ArealRadius[CTau[], CRho[]]
(2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] - ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^{2} (-2 (4 CouplingC1 + CouplingC2)
MetricA$^{(0,2)}$[CTau[], CRho[]] - 16 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} -
4 FuncG4′[KineticXEScalar[]] (4 ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} +

3 ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2) + 2 CouplingC1 MetricA$^{(1,0)}$[
CTau[], CRho[]]^2 + 5 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
12 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - 8 CouplingC1
MetricA$^{(2,0)}$[CTau[], CRho[]] + 4 CouplingC2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA$^{(2,0)}$[CTau[], CRho[]]))) +
4 ArealRadius[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(14 (4 CouplingC1 + CouplingC2)
ArealRadius[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]^3 +
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(0,1)}$[CTau[], CRho[]]
(2 (4 CouplingC1 + CouplingC2) (-1 + 3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2) -
2 ArealRadius[CTau[], CRho[]] (3 (4 CouplingC1 + CouplingC2) ArealRadius$^{(0,2)}$[
CTau[], CRho[]] + (20 CouplingC1 + 9 CouplingC2 + 16 CouplingC3)
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
(12 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
ArealRadius$^{(2,0)}$[CTau[], CRho[]]) + MetricA[CTau[], CRho[]]^2
(-2 (4 CouplingC1 + CouplingC2) + 6 (4 CouplingC1 + CouplingC2)
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 - 4 ArealRadius[CTau[], CRho[]]^2
FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 +
2 (12 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
ArealRadius[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^2 (-52 CouplingC1 MetricA$^{(0,2)}$[CTau[], CRho[]] -
13 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]] - 8 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 (ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 +
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2) - 4 FuncG4′[KineticXEScalar[]]
(3 ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 + ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2) +
14 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
8 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
18 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 4 CouplingC1
MetricA$^{(2,0)}$[CTau[], CRho[]] + 6 CouplingC2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
20 CouplingC3 MetricA$^{(2,0)}$[CTau[], CRho[]]) - 2 MetricA[CTau[], CRho[]]
(2 (4 CouplingC1 + CouplingC2) (-1 + 3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2) -
ArealRadius[CTau[], CRho[]] (3 (4 CouplingC1 + CouplingC2)
ArealRadius$^{(0,2)}$[CTau[], CRho[]] + (20 CouplingC1 + 9 CouplingC2 +
16 CouplingC3) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] - 2 (12 CouplingC1 + 3 CouplingC2 +
4 CouplingC3) ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^2 (-4 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 -
2 FuncG4′[KineticXEScalar[]] (3 ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 +
2 ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2) + (2 CouplingC1 +
3 CouplingC2 + 10 CouplingC3) MetricA$^{(2,0)}$[CTau[], CRho[]]))) +
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^2
(-2 (-1 + MetricA[CTau[], CRho[]])
(2 (4 CouplingC1 + CouplingC2) ArealRadius$^{(0,3)}$[CTau[], CRho[]] -
8 (-1 + MetricA[CTau[], CRho[]]) FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]

ClockProfile$^{(1,0)}$[CTau[], CRho[]] + 8 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} - 8 MetricA[CTau[], CRho[]]
FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} -
4 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
5 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
8 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] + 8 CouplingC1 ArealRadius$^{(1,0)}$[
CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] + 4 CouplingC2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
2 CouplingC2 ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]] (8 CouplingC1 MetricA$^{(0,3)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA$^{(0,3)}$[CTau[], CRho[]] - 8 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2})
(-2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] + ClockProfile$^{(1,0)}$[
CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) -
12 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
7 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
8 (-1 + MetricA[CTau[], CRho[]]) FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] (-2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,0)}$[CTau[], CRho[]]) -
8 CouplingC1 MetricA$^{(2,1)}$[CTau[], CRho[]] - 6 CouplingC2 MetricA$^{(2,1)}$[
CTau[], CRho[]] - 16 CouplingC3 MetricA$^{(2,1)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]]))) -
4 ArealRadius[CTau[], CRho[]]^{2} (-1 + MetricA[CTau[], CRho[]])^{3}
(4
(-1 + MetricA[CTau[], CRho[]])^{2}
FuncG4′[KineticXEScalar[]]
(-1 + 3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2})
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} +
8 (-1 + MetricA[CTau[], CRho[]])^{2}
FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{4} +
48 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]

$\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]] +$
$20\ \text{CouplingC2}\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]] +$
$32\ \text{CouplingC3}\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]] -$
$2\ \text{CouplingC1}\ \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2 -$
$18\ \text{CouplingC1}$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2$
$\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2 -$
$7\ \text{CouplingC2}\ \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2$
$\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2 -$
$8\ \text{CouplingC3}\ \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2$
$\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2 +$
$24\ \text{CouplingC1}\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,1)}[\text{CTau}[], \text{CRho}[]] +$
$10\ \text{CouplingC2}\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,1)}[\text{CTau}[], \text{CRho}[]] +$
$16\ \text{CouplingC3}\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,1)}[\text{CTau}[], \text{CRho}[]] +$
$16\ \text{CouplingC1}\ \text{ArealRadius}^{(1,1)}[\text{CTau}[], \text{CRho}[]]^2 -$
$16\ \text{CouplingC1}$
$\text{MetricA}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,1)}[\text{CTau}[], \text{CRho}[]]^2 +$
$48\ \text{CouplingC1}\ \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,2)}[\text{CTau}[], \text{CRho}[]] +$
$20\ \text{CouplingC2}\ \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,2)}[\text{CTau}[], \text{CRho}[]] +$
$32\ \text{CouplingC3}\ \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,2)}[\text{CTau}[], \text{CRho}[]] -$
$48\ \text{CouplingC1}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,2)}[\text{CTau}[], \text{CRho}[]] -$
$20\ \text{CouplingC2}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,2)}[\text{CTau}[], \text{CRho}[]] -$
$32\ \text{CouplingC3}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(1,2)}[\text{CTau}[], \text{CRho}[]] -$
$48\ \text{CouplingC1}\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(2,0)}[\text{CTau}[], \text{CRho}[]] -$
$12\ \text{CouplingC2}\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]$
$\text{ArealRadius}^{(2,0)}[\text{CTau}[], \text{CRho}[]] -$

ArealRadius[CTau[], CRho[]]
16 CouplingC3 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
48 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
4 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
32 CouplingC1 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
12 CouplingC2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
16 CouplingC3 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
64 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
24 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
32 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
32 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
12 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
16 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
4 CouplingC1 MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] -
12 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] -
4 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] +
12 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2

MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}
MetricA$^{(2,0)}$[CTau[], CRho[]] +
32 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
12 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
16 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
64 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
24 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^{4} (8 (-1 + MetricA[CTau[], CRho[]])^{5}
FuncK′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} -
4 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
8 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
4 MetricA[CTau[], CRho[]]^{2} FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
56 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
28 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2}
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
56 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
16 CouplingC1 MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
8 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
16 CouplingC3 MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
16 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
8 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
16 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
6 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} +
3 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} +

3 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} -
6 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} -
6 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} -
3 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} -
6 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} -
16 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
48 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
48 MetricA[CTau[], CRho[]]^{2} FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
16 MetricA[CTau[], CRho[]]^{3} FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
16 FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]]^{2} +
64 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]]^{2} -
96 MetricA[CTau[], CRho[]]^{2} FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]]^{2} +
64 MetricA[CTau[], CRho[]]^{3} FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]]^{2} -
16 MetricA[CTau[], CRho[]]^{4} FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]]^{2} -
72 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] -
72 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] +
72 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
72 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} -
8 CouplingC2 MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} -
16 CouplingC3 MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} +
32 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} +
16 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} +
32 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} -
16 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} -
8 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} -
16 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,1)}$[CTau[], CRho[]]^{2} -
16 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]]

16 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
16 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] +
32 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] +
8 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
24 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
24 MetricA[CTau[], CRho[]]^2 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
8 MetricA[CTau[], CRho[]]^3 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
16 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
64 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
96 MetricA[CTau[], CRho[]]^2 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
64 MetricA[CTau[], CRho[]]^3 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
16 MetricA[CTau[], CRho[]]^4 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
8 FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
32 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
48 MetricA[CTau[], CRho[]]^2 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
32 MetricA[CTau[], CRho[]]^3 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -

8 MetricA[CTau[], CRho[]]^4 FuncG4$'$[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
40 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
20 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
40 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
40 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
20 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] +
40 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
16 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] +
32 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC1 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 -
4 CouplingC2 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 -
8 CouplingC3 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 +
24 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]]^2 +

12 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]]^{2} +
24 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]]^{2} -
24 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]]^{2} -
12 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]]^{2} -
24 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]]^{2} +
8 CouplingC1 MetricA[CTau[], CRho[]]^{3} MetricA$^{(2,0)}$[CTau[], CRho[]]^{2} +
4 CouplingC2 MetricA[CTau[], CRho[]]^{3} MetricA$^{(2,0)}$[CTau[], CRho[]]^{2} +
8 CouplingC3 MetricA[CTau[], CRho[]]^{3} MetricA$^{(2,0)}$[CTau[], CRho[]]^{2} +
2 FuncG4[KineticXEScalar[]] (-1 + MetricA[CTau[], CRho[]])^{3}
(MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
2 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) -
40 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -
20 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -
40 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
80 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
40 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] +
80 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] -
40 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] -
20 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] -
40 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA$^{(2,2)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA$^{(2,2)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA$^{(2,2)}$[CTau[], CRho[]] +
48 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(2,2)}$[CTau[], CRho[]] +
24 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(2,2)}$[CTau[], CRho[]] +
48 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(2,2)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(2,2)}$[CTau[], CRho[]] -
24 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(2,2)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(2,2)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]]^{3} MetricA$^{(2,2)}$[CTau[], CRho[]] +
8 CouplingC2 MetricA[CTau[], CRho[]]^{3} MetricA$^{(2,2)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]^{3} MetricA$^{(2,2)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
24 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 24 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +

24 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] +
24 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 8 CouplingC1 MetricA[CTau[], CRho[]]^{3}
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
4 CouplingC2 MetricA[CTau[], CRho[]]^{3} MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA[CTau[], CRho[]]^{3}
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]]) -
4 ArealRadius[CTau[], CRho[]]^{3} (-1 + MetricA[CTau[], CRho[]])
((4 CouplingC1 + CouplingC2) MetricA$^{(0,1)}$[CTau[], CRho[]]^{2}
(19 ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
5 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
4 (-1 + MetricA[CTau[], CRho[]])^{2} FuncG4′[KineticXEScalar[]]
(2 ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} (2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(-4 ClockProfile$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] -
2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(3 ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) +
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(0,1)}$[CTau[], CRho[]]
(-12 (4 CouplingC1 + CouplingC2) ArealRadius$^{(0,3)}$[CTau[], CRho[]] -
8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} - (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}) + (4 CouplingC1 + CouplingC2)
(6 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
5 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(2,1)}$[CTau[], CRho[]])) +
(-1 + MetricA[CTau[], CRho[]]) (-16 CouplingC1 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
4 CouplingC2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
16 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ClockProfile$^{(0,2)}$[
CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[],
CRho[]] + 16 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
16 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} +
32 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,2)}$[CTau[],

CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} -
16 MetricA[CTau[], CRho[]]^{2} FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,2)}$[CTau[],
CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} +
8 CouplingC1 MetricA$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 2 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[
CTau[], CRho[]] - 8 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{4} MetricA$^{(1,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[],
CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{4} MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 MetricA[CTau[], CRho[]]^{2} FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[],
CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{4} MetricA$^{(1,0)}$[CTau[], CRho[]] +
6 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} +
CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} -
2 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} +
16 CouplingC1 ArealRadius$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
4 CouplingC2 ArealRadius$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] - 4 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] - 4 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
8 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
2 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
8 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] - 2 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
4 FuncG4[KineticXEScalar[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
12 FuncG4[KineticXEScalar[]] MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[
CTau[], CRho[]] - 12 FuncG4[KineticXEScalar[]] MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 4 FuncG4[KineticXEScalar[]]
MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
12 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
3 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
12 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 3 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
16 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]

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      MetricA^(1,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] +
    3 CouplingC2 ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]
      MetricA^(2,0)[CTau[], CRho[]] - 4 CouplingC3 ArealRadius^(1,0)[CTau[], CRho[]]
      MetricA^(1,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] -
    16 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius^(1,0)[CTau[], CRho[]]
      MetricA^(1,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] -
    3 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius^(1,0)[CTau[], CRho[]]
      MetricA^(1,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] +
    4 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius^(1,0)[CTau[], CRho[]]
      MetricA^(1,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] -
    16 CouplingC1 ArealRadius^(2,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] -
    4 CouplingC2 ArealRadius^(2,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] +
    32 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]]
      MetricA^(2,0)[CTau[], CRho[]] + 8 CouplingC2 MetricA[CTau[], CRho[]]
      ArealRadius^(2,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] -
    16 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius^(2,0)[CTau[], CRho[]]
      MetricA^(2,0)[CTau[], CRho[]] - 4 CouplingC2 MetricA[CTau[], CRho[]]^2
      ArealRadius^(2,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] +
    ArealRadius^(0,2)[CTau[], CRho[]] (-8 (4 CouplingC1 + CouplingC2) MetricA^(0,2)[
            CTau[], CRho[]] + 2 (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
          MetricA^(1,0)[CTau[], CRho[]]^2 + 8 (CouplingC1 - CouplingC3)
          (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,0)[CTau[], CRho[]]) -
    16 CouplingC1 ArealRadius^(2,2)[CTau[], CRho[]] -
    4 CouplingC2 ArealRadius^(2,2)[CTau[], CRho[]] +
    32 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius^(2,2)[CTau[], CRho[]] +
    8 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius^(2,2)[CTau[], CRho[]] -
    16 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius^(2,2)[CTau[], CRho[]] -
    4 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius^(2,2)[CTau[], CRho[]] +
    8 CouplingC1 MetricA^(1,0)[CTau[], CRho[]] ArealRadius^(3,0)[CTau[], CRho[]] +
    2 CouplingC2 MetricA^(1,0)[CTau[], CRho[]] ArealRadius^(3,0)[CTau[], CRho[]] -
    16 CouplingC1 MetricA[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]
      ArealRadius^(3,0)[CTau[], CRho[]] - 4 CouplingC2 MetricA[CTau[], CRho[]]
      MetricA^(1,0)[CTau[], CRho[]] ArealRadius^(3,0)[CTau[], CRho[]] +
    8 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA^(1,0)[CTau[], CRho[]]
      ArealRadius^(3,0)[CTau[], CRho[]] + 2 CouplingC2 MetricA[CTau[], CRho[]]^2
      MetricA^(1,0)[CTau[], CRho[]] ArealRadius^(3,0)[CTau[], CRho[]] +
    8 CouplingC1 ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(3,0)[CTau[], CRho[]] +
    2 CouplingC2 ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(3,0)[CTau[], CRho[]] -
    16 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius^(1,0)[CTau[], CRho[]]
      MetricA^(3,0)[CTau[], CRho[]] - 4 CouplingC2 MetricA[CTau[], CRho[]]
      ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(3,0)[CTau[], CRho[]] +
    8 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius^(1,0)[CTau[], CRho[]]
      MetricA^(3,0)[CTau[], CRho[]] + 2 CouplingC2 MetricA[CTau[], CRho[]]^2
      ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(3,0)[CTau[], CRho[]]))) /
 (8 ArealRadius[CTau[], CRho[]]^4 (-1 + MetricA[CTau[], CRho[]])^5) ==
0
```

```
SphericalEquations["EuclideanMetricEOM"][{"CTau", "CRho"}]
```

```
(-4 (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius^(0,1)[CTau[], CRho[]]^3
      (4 (3 CouplingC1 + CouplingC2 + CouplingC3) (-1 + MetricA[CTau[], CRho[]])
          ArealRadius^(1,0)[CTau[], CRho[]] + (4 CouplingC1 + CouplingC2)
          ArealRadius[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]) +
    ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(0,1)[CTau[], CRho[]]^2
      (ArealRadius[CTau[], CRho[]] (4 (-1 + MetricA[CTau[], CRho[]])^2 FuncG4′[KineticXEScalar[]]
                ClockProfile^(0,1)[CTau[], CRho[]] ClockProfile^(1,0)[CTau[], CRho[]] -
              8 (-1 + MetricA[CTau[], CRho[]]) FuncG4″[KineticXEScalar[]]
                ClockProfile^(0,1)[CTau[], CRho[]]^3 ClockProfile^(1,0)[CTau[], CRho[]] -
              (8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3) MetricA^(0,1)[CTau[], CRho[]]
                MetricA^(1,0)[CTau[], CRho[]]) + 8 (4 CouplingC1 + CouplingC2)
          (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius^(1,1)[CTau[], CRho[]]) +
    ArealRadius^(0,1)[CTau[], CRho[]] (16 (3 CouplingC1 + CouplingC2 + CouplingC3)
          (-1 + MetricA[CTau[], CRho[]])^4 ArealRadius^(1,0)[CTau[], CRho[]]^3 +
        4 (4 CouplingC1 + CouplingC2) ArealRadius[CTau[], CRho[]]
          (-1 + MetricA[CTau[], CRho[]])^3
          ArealRadius^(1,0)[CTau[], CRho[]]^2 MetricA^(1,0)[CTau[], CRho[]] -
        (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(1,0)[CTau[], CRho[]]
          (-16 (3 CouplingC1 + CouplingC2 + CouplingC3) + 16 (3 CouplingC1 + CouplingC2 +
                    CouplingC3) MetricA[CTau[], CRho[]]^3 - ArealRadius[CTau[], CRho[]]^2
              (8 (2 CouplingC1 + CouplingC2 + 2 CouplingC3) MetricA^(0,1)[CTau[], CRho[]]^2 +
                  4 (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
                    MetricA^(0,2)[CTau[], CRho[]] + 16 FuncG4″[KineticXEScalar[]]
                    ClockProfile^(0,1)[CTau[], CRho[]]^2 ClockProfile^(1,0)[CTau[], CRho[]]^2 -
                  4 CouplingC1 MetricA^(1,0)[CTau[], CRho[]]^2 - CouplingC2
                    MetricA^(1,0)[CTau[], CRho[]]^2 - 8 CouplingC1 MetricA^(2,0)[CTau[], CRho[]] +
                  8 CouplingC3 MetricA^(2,0)[CTau[], CRho[]]) + 8 MetricA[CTau[], CRho[]]^2
              (-6 (3 CouplingC1 + CouplingC2 + CouplingC3) + ArealRadius[CTau[], CRho[]]^2
                    (-2 FuncG4″[KineticXEScalar[]] ClockProfile^(0,1)[CTau[], CRho[]]^2
                          ClockProfile^(1,0)[CTau[], CRho[]]^2 + (CouplingC1 - CouplingC3)
                          MetricA^(2,0)[CTau[], CRho[]])) + MetricA[CTau[], CRho[]]
              (48 (3 CouplingC1 + CouplingC2 + CouplingC3) + ArealRadius[CTau[], CRho[]]^2
                    (4 (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
                          MetricA^(0,2)[CTau[], CRho[]] + 32 FuncG4″[KineticXEScalar[]]
                          ClockProfile^(0,1)[CTau[], CRho[]]^2 ClockProfile^(1,0)[CTau[], CRho[]]^2 -
                      4 CouplingC1 MetricA^(1,0)[CTau[], CRho[]]^2 -
                      CouplingC2 MetricA^(1,0)[CTau[], CRho[]]^2 - 16 CouplingC1 MetricA^(2,0)[
                            CTau[], CRho[]] + 16 CouplingC3 MetricA^(2,0)[CTau[], CRho[]]))) +
        ArealRadius[CTau[], CRho[]]^2 (2 (-1 + MetricA[CTau[], CRho[]])^2
              ((8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
                    ArealRadius^(0,2)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] +
                  (8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3) MetricA^(0,1)[CTau[], CRho[]]
                    ArealRadius^(1,1)[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
                    ((8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
                          MetricA^(1,0)[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]] +
```

4 (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(3,0)}$[CTau[], CRho[]])) +
ArealRadius[CTau[], CRho[]] (2 FuncG4[KineticXEScalar[]]
(-1 + MetricA[CTau[], CRho[]])^{3} MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{4} MetricA$^{(1,0)}$[
CTau[], CRho[]] - 8 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{4} MetricA$^{(1,0)}$[CTau[], CRho[]] +
56 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] +
14 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] +
20 CouplingC1 MetricA$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
5 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
20 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] - 5 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}
MetricA$^{(1,0)}$[CTau[], CRho[]] + 8 MetricA[CTau[], CRho[]]^{2}
FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} - 3 CouplingC2 MetricA$^{(1,0)}$[
CTau[], CRho[]]^{3} - 10 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} +
2 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} +
3 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} +
10 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} +
16 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{3}
ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
32 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{3} ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
16 MetricA[CTau[], CRho[]]^{2} FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{3} ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
16 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
48 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,1)}$[CTau[], CRho[]] + 48 MetricA[CTau[], CRho[]]^{2}
FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]]^{3} FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
4 (-1 + MetricA[CTau[], CRho[]])^{2} FuncG4′[KineticXEScalar[]]
(2 ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}
MetricA$^{(1,0)}$[CTau[], CRho[]] - 2 (-1 + MetricA[CTau[], CRho[]])

ClockProfile$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]]) +
32 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
8 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] - 8 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] + 8 CouplingC1
MetricA$^{(1,2)}$[CTau[], CRho[]] + 2 CouplingC2 MetricA$^{(1,2)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
4 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,2)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,2)}$[CTau[], CRho[]] -
4 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
20 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 12 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
40 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 4 CouplingC1 MetricA[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 20 CouplingC3 MetricA[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] - 2 CouplingC2
MetricA$^{(3,0)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA$^{(3,0)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
24 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(3,0)}$[CTau[], CRho[]] -
24 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(3,0)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA[CTau[], CRho[]]^3 MetricA$^{(3,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]]^3 MetricA$^{(3,0)}$[CTau[], CRho[]]))) -
ArealRadius[CTau[], CRho[]] (8 (4 CouplingC1 + CouplingC2)
(-1 + MetricA[CTau[], CRho[]])^4
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,1)}$[
CTau[], CRho[]] -
2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^2
(2 (2 CouplingC1 + CouplingC2 + 2 CouplingC3) MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] (-3 ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) (4 (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
ArealRadius$^{(0,3)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2)
ClockProfile$^{(1,0)}$[CTau[], CRho[]] + 4 FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3 - 4 MetricA[CTau[], CRho[]]

FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} -
16 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[
CTau[], CRho[]] - 6 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] - 8 CouplingC3
ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
4 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] - 3 CouplingC2 ArealRadius$^{(1,0)}$[CTau[],
CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] - 4 CouplingC1
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,1)}$[CTau[], CRho[]] -
2 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,1)}$[
CTau[], CRho[]] - 4 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}
MetricA$^{(1,1)}$[CTau[], CRho[]] - 16 CouplingC1
ArealRadius$^{(1,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
6 CouplingC2 ArealRadius$^{(1,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[
CTau[], CRho[]] - 8 CouplingC3 ArealRadius$^{(1,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 16 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 8 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]])) +
ArealRadius[CTau[], CRho[]]^{2} (-1 + MetricA[CTau[], CRho[]])
(4 (4 CouplingC1 + CouplingC2) MetricA$^{(0,1)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
MetricA$^{(0,1)}$[CTau[], CRho[]] (15 (4 CouplingC1 + CouplingC2)
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (4 CouplingC1 + CouplingC2) (-1 + MetricA[CTau[], CRho[]])
(3 ArealRadius$^{(1,2)}$[CTau[], CRho[]] - MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]]) - ArealRadius$^{(1,0)}$[CTau[], CRho[]]
((20 CouplingC1 + 7 CouplingC2 + 8 CouplingC3) MetricA$^{(1,0)}$[CTau[],
CRho[]]^{2} - 4 (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]])) -
(-1 + MetricA[CTau[], CRho[]]) (6 (4 CouplingC1 + CouplingC2)
ArealRadius$^{(0,3)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{3}
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 8 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} MetricA$^{(1,0)}$[CTau[], CRho[]] -
8 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} MetricA$^{(1,0)}$[CTau[], CRho[]] -
4 FuncG4[KineticXEScalar[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
8 FuncG4[KineticXEScalar[]] MetricA[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] - 4 FuncG4[KineticXEScalar[]]

ArealRadius$^{(1,1)}$[CTau[], CRho[]] - 4 FuncG4[KineticXEScalar[]]
MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,1)}$[CTau[], CRho[]] + 8 CouplingC1
MetricA$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(1,1)}$[CTau[], CRho[]] - 5 CouplingC2
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
4 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(1,1)}$[CTau[], CRho[]] + 16 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
16 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
32 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3}
ClockProfile$^{(1,1)}$[CTau[], CRho[]] + 16 MetricA[CTau[], CRho[]]^{2}
FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
4 (-1 + MetricA[CTau[], CRho[]]) FuncG4′[KineticXEScalar[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]])) + 24 CouplingC1
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
6 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
20 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
7 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
16 CouplingC1 ArealRadius$^{(1,3)}$[CTau[], CRho[]] +
4 CouplingC2 ArealRadius$^{(1,3)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,3)}$[CTau[], CRho[]] -
4 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,3)}$[CTau[], CRho[]] -
8 CouplingC1 MetricA$^{(1,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
2 CouplingC2 MetricA$^{(1,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 8 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 16 CouplingC1

```
            ArealRadius      [CTau[], CRho[]] - 16 CouplingC1
              ArealRadius^(1,1)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] -
            6 CouplingC2 ArealRadius^(1,1)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] -
            8 CouplingC3 ArealRadius^(1,1)[CTau[], CRho[]]
              MetricA^(2,0)[CTau[], CRho[]] + 16 CouplingC1 MetricA[CTau[], CRho[]]
              ArealRadius^(1,1)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] +
            6 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius^(1,1)[CTau[], CRho[]]
              MetricA^(2,0)[CTau[], CRho[]] + 8 CouplingC3 MetricA[CTau[], CRho[]]
              ArealRadius^(1,1)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] -
            8 CouplingC1 ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(2,1)[CTau[], CRho[]] -
            4 CouplingC2 ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(2,1)[CTau[], CRho[]] -
            8 CouplingC3 ArealRadius^(1,0)[CTau[], CRho[]]
              MetricA^(2,1)[CTau[], CRho[]] + 8 CouplingC1 MetricA[CTau[], CRho[]]
              ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(2,1)[CTau[], CRho[]] +
            4 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius^(1,0)[CTau[], CRho[]]
              MetricA^(2,1)[CTau[], CRho[]] + 8 CouplingC3 MetricA[CTau[], CRho[]]
              ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(2,1)[CTau[], CRho[]] +
            16 CouplingC1 ArealRadius^(3,1)[CTau[], CRho[]] +
            4 CouplingC2 ArealRadius^(3,1)[CTau[], CRho[]] -
            32 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius^(3,1)[CTau[], CRho[]] -
            8 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius^(3,1)[CTau[], CRho[]] +
            16 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius^(3,1)[CTau[], CRho[]] +
            4 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius^(3,1)[CTau[], CRho[]])) +
      ArealRadius[CTau[], CRho[]]^3 (-2 (-1 + MetricA[CTau[], CRho[]])^4
          FuncK′[KineticXEScalar[]] ClockProfile^(0,1)[CTau[], CRho[]]
          ClockProfile^(1,0)[CTau[], CRho[]] +
        (2 CouplingC1 + CouplingC2 + 2 CouplingC3)
          (MetricA^(0,1)[CTau[], CRho[]]
              (7 MetricA^(1,0)[CTau[], CRho[]]^3 - 9 (-1 + MetricA[CTau[], CRho[]])
                  MetricA^(1,0)[CTau[], CRho[]] MetricA^(2,0)[CTau[], CRho[]] +
                2 (-1 + MetricA[CTau[], CRho[]])^2 MetricA^(3,0)[CTau[], CRho[]]) -
            (-1 + MetricA[CTau[], CRho[]]) (7 MetricA^(1,0)[CTau[], CRho[]]^2
                  MetricA^(1,1)[CTau[], CRho[]] - 5 (-1 + MetricA[CTau[], CRho[]])
                  MetricA^(1,0)[CTau[], CRho[]] MetricA^(2,1)[CTau[], CRho[]] +
                2 (-1 + MetricA[CTau[], CRho[]]) (-2 MetricA^(1,1)[CTau[], CRho[]]
                    MetricA^(2,0)[CTau[], CRho[]] + (-1 + MetricA[CTau[], CRho[]])
                    MetricA^(3,1)[CTau[], CRho[]])))))) /
  (2 ArealRadius[CTau[], CRho[]]^4 (-1 + MetricA[CTau[], CRho[]])^4) ==
 0
```

```
SphericalEquations["EuclideanMetricEOM"][{"CRho", "CRho"}]
```

```
(-16 (8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
      (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius^(0,1)[CTau[], CRho[]]^4 +
    8 (4 CouplingC1 + CouplingC2) ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
      ArealRadius^(0,1)[CTau[], CRho[]]^3 MetricA^(0,1)[CTau[], CRho[]] +
    16 (4 CouplingC1 + CouplingC2) (-1 + MetricA[CTau[], CRho[]])^4
      ArealRadius^(1,0)[CTau[], CRho[]]^2 (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2) +
```

$$
\begin{aligned}
&8\,(4\,\text{CouplingC1}+\text{CouplingC2})\,\text{ArealRadius}[\text{CTau}[],\text{CRho}[]]\,(-1+\text{MetricA}[\text{CTau}[],\text{CRho}[]])^3\\
&\quad(2\,\text{ArealRadius}^{(0,2)}[\text{CTau}[],\text{CRho}[]]\,(-1+\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2)+\\
&\qquad\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]\,\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]-\\
&\qquad\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^3\,\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]+\\
&\qquad 2\,(-1+\text{MetricA}[\text{CTau}[],\text{CRho}[]])\,\text{ArealRadius}^{(2,0)}[\text{CTau}[],\text{CRho}[]]-\\
&\qquad 6\,(-1+\text{MetricA}[\text{CTau}[],\text{CRho}[]])\,\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2\\
&\qquad\quad\text{ArealRadius}^{(2,0)}[\text{CTau}[],\text{CRho}[]])+\\
&4\,\text{ArealRadius}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^2\,(4\,(8\,\text{CouplingC1}+3\,\text{CouplingC2}+4\,\text{CouplingC3}-\\
&\qquad\qquad 2\,(2\,\text{CouplingC1}+\text{CouplingC2}+2\,\text{CouplingC3})\,\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2)-\\
&\qquad 2\,(4\,\text{CouplingC1}+\text{CouplingC2})\,\text{ArealRadius}[\text{CTau}[],\text{CRho}[]]\\
&\qquad\quad(2\,\text{ArealRadius}^{(0,2)}[\text{CTau}[],\text{CRho}[]]-\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]\\
&\qquad\qquad\quad\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]+6\,\text{ArealRadius}^{(2,0)}[\text{CTau}[],\text{CRho}[]])+\\
&\qquad 4\,\text{MetricA}[\text{CTau}[],\text{CRho}[]]^3\,(-8\,\text{CouplingC1}-3\,\text{CouplingC2}-4\,\text{CouplingC3}+\\
&\qquad\qquad 2\,(2\,\text{CouplingC1}+\text{CouplingC2}+2\,\text{CouplingC3})\,\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2+\\
&\qquad\qquad 3\,(4\,\text{CouplingC1}+\text{CouplingC2})\,\text{ArealRadius}[\text{CTau}[],\text{CRho}[]]\\
&\qquad\qquad\quad\text{ArealRadius}^{(2,0)}[\text{CTau}[],\text{CRho}[]])+\text{ArealRadius}[\text{CTau}[],\text{CRho}[]]^2\\
&\qquad\quad(12\,\text{FuncG4}'[\text{KineticXEScalar}[]]\,\text{ClockProfile}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^2+\\
&\qquad\qquad 8\,\text{FuncG4}''[\text{KineticXEScalar}[]]\,\text{ClockProfile}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^4+24\,\text{CouplingC1}\\
&\qquad\qquad\quad\text{MetricA}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^2+9\,\text{CouplingC2}\,\text{MetricA}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^2+\\
&\qquad\qquad 12\,\text{CouplingC3}\,\text{MetricA}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^2+16\,\text{CouplingC1}\\
&\qquad\qquad\quad\text{MetricA}^{(0,2)}[\text{CTau}[],\text{CRho}[]]+6\,\text{CouplingC2}\,\text{MetricA}^{(0,2)}[\text{CTau}[],\text{CRho}[]]+\\
&\qquad\qquad 8\,\text{CouplingC3}\,\text{MetricA}^{(0,2)}[\text{CTau}[],\text{CRho}[]]+18\,\text{CouplingC1}\\
&\qquad\qquad\quad\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2+7\,\text{CouplingC2}\,\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2+\\
&\qquad\qquad 16\,\text{CouplingC3}\,\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2+12\,\text{CouplingC1}\\
&\qquad\qquad\quad\text{MetricA}^{(2,0)}[\text{CTau}[],\text{CRho}[]]+6\,\text{CouplingC2}\,\text{MetricA}^{(2,0)}[\text{CTau}[],\text{CRho}[]]+\\
&\qquad\qquad 16\,\text{CouplingC3}\,\text{MetricA}^{(2,0)}[\text{CTau}[],\text{CRho}[]])-\text{MetricA}[\text{CTau}[],\text{CRho}[]]\\
&\qquad\quad(12\,(8\,\text{CouplingC1}+3\,\text{CouplingC2}+4\,\text{CouplingC3}-2\,(2\,\text{CouplingC1}+\\
&\qquad\qquad\qquad\quad\text{CouplingC2}+2\,\text{CouplingC3})\,\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2)-\\
&\qquad\qquad 4\,(4\,\text{CouplingC1}+\text{CouplingC2})\,\text{ArealRadius}[\text{CTau}[],\text{CRho}[]]\\
&\qquad\qquad\quad(2\,\text{ArealRadius}^{(0,2)}[\text{CTau}[],\text{CRho}[]]-\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]\\
&\qquad\qquad\qquad\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]+9\,\text{ArealRadius}^{(2,0)}[\text{CTau}[],\text{CRho}[]])+\\
&\qquad\qquad\text{ArealRadius}[\text{CTau}[],\text{CRho}[]]^2\,(24\,\text{FuncG4}'[\text{KineticXEScalar}[]]\\
&\qquad\qquad\qquad\text{ClockProfile}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^2+8\,\text{FuncG4}''[\text{KineticXEScalar}[]]\\
&\qquad\qquad\qquad\text{ClockProfile}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^4+16\,\text{CouplingC1}\\
&\qquad\qquad\qquad\text{MetricA}^{(0,2)}[\text{CTau}[],\text{CRho}[]]+6\,\text{CouplingC2}\,\text{MetricA}^{(0,2)}[\text{CTau}[],\text{CRho}[]]+\\
&\qquad\qquad\quad 8\,\text{CouplingC3}\,\text{MetricA}^{(0,2)}[\text{CTau}[],\text{CRho}[]]+18\,\text{CouplingC1}\,\text{MetricA}^{(1,0)}[\\
&\qquad\qquad\qquad\quad\text{CTau}[],\text{CRho}[]]^2+7\,\text{CouplingC2}\,\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2+\\
&\qquad\qquad\quad 16\,\text{CouplingC3}\,\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2+24\,\text{CouplingC1}\,\text{MetricA}^{(2,0)}[\\
&\qquad\qquad\qquad\text{CTau}[],\text{CRho}[]]+12\,\text{CouplingC2}\,\text{MetricA}^{(2,0)}[\text{CTau}[],\text{CRho}[]]+\\
&\qquad\qquad\quad 32\,\text{CouplingC3}\,\text{MetricA}^{(2,0)}[\text{CTau}[],\text{CRho}[]]))+2\,\text{MetricA}[\text{CTau}[],\text{CRho}[]]^2\\
&\qquad\quad(6\,(8\,\text{CouplingC1}+3\,\text{CouplingC2}+4\,\text{CouplingC3}-2\,(2\,\text{CouplingC1}+\\
&\qquad\qquad\qquad\quad\text{CouplingC2}+2\,\text{CouplingC3})\,\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]^2)-\\
&\qquad\qquad(4\,\text{CouplingC1}+\text{CouplingC2})\,\text{ArealRadius}[\text{CTau}[],\text{CRho}[]]\\
&\qquad\qquad\quad(2\,\text{ArealRadius}^{(0,2)}[\text{CTau}[],\text{CRho}[]]-\text{ArealRadius}^{(1,0)}[\text{CTau}[],\text{CRho}[]]\\
&\qquad\qquad\qquad\text{MetricA}^{(1,0)}[\text{CTau}[],\text{CRho}[]]+18\,\text{ArealRadius}^{(2,0)}[\text{CTau}[],\text{CRho}[]])+\\
&\qquad\qquad\text{ArealRadius}[\text{CTau}[],\text{CRho}[]]^2\,(6\,\text{FuncG4}'[\text{KineticXEScalar}[]]
\end{aligned}
$$

ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 + (6 CouplingC1 +
3 CouplingC2 + 8 CouplingC3) MetricA$^{(2,0)}$[CTau[], CRho[]]))) -
4 ArealRadius[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(MetricA$^{(0,1)}$[CTau[], CRho[]]
(2 (4 CouplingC1 + CouplingC2) (-1 + MetricA[CTau[], CRho[]])^2
(-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2) + 2 ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) ((8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
ArealRadius$^{(0,2)}$[CTau[], CRho[]] - 2 (CouplingC2 + 4 CouplingC3)
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
(12 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(2,0)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]]^2 (2 FuncG4[KineticXEScalar[]]
(-1 + MetricA[CTau[], CRho[]])^2 - 4 (-1 + MetricA[CTau[], CRho[]])
FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 +
(4 CouplingC1 + CouplingC2) (11 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
4 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]))) +
2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
(ArealRadius[CTau[], CRho[]] (3 (4 CouplingC1 + CouplingC2) MetricA$^{(1,0)}$[CTau[],
CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] - 8 (-1 + MetricA[CTau[], CRho[]])
FuncG4″[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 - (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2) ClockProfile$^{(2,0)}$[CTau[],
CRho[]] + 2 (-1 + MetricA[CTau[], CRho[]]) FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] (-2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
4 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,0)}$[CTau[], CRho[]])) -
(-1 + MetricA[CTau[], CRho[]]) (2 (8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
ArealRadius$^{(0,3)}$[CTau[], CRho[]] - 8 (-1 + MetricA[CTau[], CRho[]])
FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
8 FuncG4″[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
4 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
8 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
4 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[],
CRho[]] + 24 CouplingC1 ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
10 CouplingC2 ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
16 CouplingC3 ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
24 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
10 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]]))) -
8 ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^2
(2 (8 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)

(2 (3 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 -
4 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 +
4 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 +
2 (-1 + MetricA[CTau[], CRho[]]) FuncG4′[KineticXEScalar[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2) -
2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2) +
CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 3 CouplingC1
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
4 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
8 CouplingC1 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 +
8 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 +
2 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
2 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
12 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
4 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
12 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]

8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius[illegible][CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
(2 (4 CouplingC1 + CouplingC2) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + (12 CouplingC1 + 3 CouplingC2 + 4 CouplingC3)
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
2 CouplingC1 MetricA$^{(2,0)}$[CTau[], CRho[]] - 2 CouplingC1
MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
2 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
2 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
2 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
8 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
2 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]]) +
4 ArealRadius[CTau[], CRho[]]^3 (-1 + MetricA[CTau[], CRho[]])
(-40 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
10 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
28 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^3 +
11 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 +
16 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^3 -
2 FuncG4[KineticXEScalar[]] (-1 + MetricA[CTau[], CRho[]])^2
(2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) -
32 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
16 CouplingC1 ArealRadius$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
4 CouplingC2 ArealRadius$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +

32 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
8 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
12 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
12 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
4 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
4 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
16 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
32 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]]^2 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
16 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3 ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
48 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
48 MetricA[CTau[], CRho[]]^2 FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
16 MetricA[CTau[], CRho[]]^3 FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
4 (-1 + MetricA[CTau[], CRho[]]) FuncG4′[KineticXEScalar[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[],
CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] - 2 (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(-2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
4 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) -
32 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]]

ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] +
48 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
20 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
32 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
20 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC1 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
2 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
16 CouplingC1 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
4 CouplingC2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
8 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
4 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
4 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]

4 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
2 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
16 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
8 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
16 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
16 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 32 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] +
8 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] + 16 CouplingC3 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
16 CouplingC1 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
4 CouplingC2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
48 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
4 CouplingC2 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(4,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^{4} (8 (-1 + MetricA[CTau[], CRho[]])^{3}
FuncK′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} -
4 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
4 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
50 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} -
25 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} -
50 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{4} -
16 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
32 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]]^{2} FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
16 FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]]^{2} +
48 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]]^{2} - 48 MetricA[CTau[], CRho[]]^{2}
FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]]^{2} +
16 MetricA[CTau[], CRho[]]^{3} FuncG4′[KineticXEScalar[]]

16 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]]^2 +
8 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
8 MetricA[CTau[], CRho[]]^2 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
16 FuncG4′[KineticXEScalar[]] ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
48 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
48 MetricA[CTau[], CRho[]]^2 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]]^3 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
8 FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
24 MetricA[CTau[], CRho[]] FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
24 MetricA[CTau[], CRho[]]^2 FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
8 MetricA[CTau[], CRho[]]^3 FuncG4′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
120 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
60 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
120 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
120 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
60 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] + 120 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
40 CouplingC1 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 -
20 CouplingC2 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 -
40 CouplingC3 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 +
80 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]]^2 +
40 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]]^2 +
80 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]]^2 -
40 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 -
20 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 -
40 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 -
2 FuncG4[KineticXEScalar[]] (-1 + MetricA[CTau[], CRho[]])^2
(MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
2 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) -
48 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
24 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]]

48 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
48 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] + 96 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 24 CouplingC2 MetricA[CTau[], CRho[]]^{2}
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 16 CouplingC1 MetricA$^{(4,0)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA$^{(4,0)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA$^{(4,0)}$[CTau[], CRho[]] +
48 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(4,0)}$[CTau[], CRho[]] +
24 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(4,0)}$[CTau[], CRho[]] +
48 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(4,0)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(4,0)}$[CTau[], CRho[]] -
24 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(4,0)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(4,0)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]]^{3} MetricA$^{(4,0)}$[CTau[], CRho[]] +
8 CouplingC2 MetricA[CTau[], CRho[]]^{3} MetricA$^{(4,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]^{3} MetricA$^{(4,0)}$[CTau[], CRho[]])) /
(8 ArealRadius[CTau[], CRho[]]^{4} (-1 + MetricA[CTau[], CRho[]])^{3}) ==
0

```
SphericalEquations["EuclideanMetricEOM"][{"CTheta", "CTheta"}]
```

-((8 (-1 + MetricA[CTau[], CRho[]])^{3}
(1 - ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} - ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} +
MetricA[CTau[], CRho[]] (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}))
(-2 CouplingC1 - CouplingC2 - 2 CouplingC3 -
2 (3 CouplingC1 + CouplingC2 + CouplingC3) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} -
6 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} -
2 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} -
2 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} + MetricA[CTau[], CRho[]]
(2 CouplingC1 + CouplingC2 + 2 CouplingC3 + 2 (3 CouplingC1 +
CouplingC2 + CouplingC3) ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2})) +
4 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^{2}
(6 (3 CouplingC1 + CouplingC2 + CouplingC3) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{3}
MetricA$^{(0,1)}$[CTau[], CRho[]] + (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(0,1)}$[CTau[], CRho[]] (MetricA$^{(0,1)}$[CTau[], CRho[]]
(10 CouplingC1 + 3 CouplingC2 + 2 CouplingC3 - 2 (3 CouplingC1 +
CouplingC2 + CouplingC3) ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}) +
16 (3 CouplingC1 + CouplingC2 + CouplingC3) (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]) -
2 (3 CouplingC1 + CouplingC2 + CouplingC3) (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} (6 ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -

$$
\begin{array}{l}
2\,(-1 + \text{MetricA}[\text{CTau}[], \text{CRho}[]])\ \text{ArealRadius}^{(2,0)}[\text{CTau}[], \text{CRho}[]]) + \\
(-1 + \text{MetricA}[\text{CTau}[], \text{CRho}[]])^2\ (2\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] \\
(-10\ \text{CouplingC1} - 3\ \text{CouplingC2} - 2\ \text{CouplingC3} + 2\ (3\ \text{CouplingC1} + \\
\text{CouplingC2} + \text{CouplingC3})\ \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2) + \\
(10\ \text{CouplingC1} + 3\ \text{CouplingC2} + 2\ \text{CouplingC3})\ \text{ArealRadius}^{(1,0)}[\\
\text{CTau}[], \text{CRho}[]]\ \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]] - \\
2\ (3\ \text{CouplingC1} + \text{CouplingC2} + \text{CouplingC3}) \\
\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^3\ \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]] + \\
2\ (10\ \text{CouplingC1} + 3\ \text{CouplingC2} + 2\ \text{CouplingC3}) \\
(-1 + \text{MetricA}[\text{CTau}[], \text{CRho}[]])\ \text{ArealRadius}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
12\ (3\ \text{CouplingC1} + \text{CouplingC2} + \text{CouplingC3})\ (-1 + \text{MetricA}[\text{CTau}[], \text{CRho}[]]) \\
\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2\ \text{ArealRadius}^{(2,0)}[\text{CTau}[], \text{CRho}[]])) + \\
\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]^2\ (-1 + \text{MetricA}[\text{CTau}[], \text{CRho}[]]) \\
(-8\ \text{CouplingC1}\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]^2\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]^2 - \\
3\ \text{CouplingC2}\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]^2\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]^2 - \\
4\ \text{CouplingC3}\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]^2\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]^2 - \\
32\ \text{CouplingC1}\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
\text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] - \\
12\ \text{CouplingC2}\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
\text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] - 16\ \text{CouplingC3}\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
\text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] + \\
32\ \text{CouplingC1}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
\text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] + \\
12\ \text{CouplingC2}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
\text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] + \\
16\ \text{CouplingC3}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
\text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] - \\
32\ \text{CouplingC1}\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 - \\
12\ \text{CouplingC2}\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 - \\
16\ \text{CouplingC3}\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 + \\
64\ \text{CouplingC1}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 + \\
24\ \text{CouplingC2}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 + \\
32\ \text{CouplingC3}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 - \\
32\ \text{CouplingC1}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 - \\
12\ \text{CouplingC2}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 - \\
16\ \text{CouplingC3}\ \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\ \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]^2 + \\
4\ \text{FuncG4}[\text{KineticXEScalar}[]]\ (-1 + \text{MetricA}[\text{CTau}[], \text{CRho}[]])^3 \\
(1 - \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]^2 - \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2 + \\
\text{MetricA}[\text{CTau}[], \text{CRho}[]]\ (-1 + \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2)) + \\
8\ (-1 + \text{MetricA}[\text{CTau}[], \text{CRho}[]])^2\ \text{FuncG4}'[\text{KineticXEScalar}[]] \\
(\text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{ClockProfile}^{(0,1)}[\text{CTau}[], \text{CRho}[]] - \\
(-1 + \text{MetricA}[\text{CTau}[], \text{CRho}[]])\ \text{ArealRadius}^{(1,0)}[\text{CTau}[], \\
\text{CRho}[]]\ \text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]])^2 + \\
16\ \text{CouplingC1}\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\ \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]] + \\
6\ \text{CouplingC2}\ \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\ \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]
\end{array}
$$

ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] - 6 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 32 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
12 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 16 CouplingC3 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
64 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
24 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
4 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
8 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
4 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
28 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
2 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
8 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
28 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
2 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 8 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
4 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
4 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
8 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 2 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 4 CouplingC1 MetricA[CTau[], CRho[]]^2

MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 4 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - 4 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
64 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] - 32 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
128 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
64 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
64 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
64 CouplingC1 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
32 CouplingC3 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
192 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 +
96 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 +
192 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
96 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
64 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 +
32 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
64 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 12 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
128 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
64 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
128 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
24 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
384 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 72 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
384 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 72 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
128 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 24 CouplingC2 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
64 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]

64 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 12 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
192 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
192 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
64 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
32 CouplingC1 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
12 CouplingC2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
16 CouplingC3 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
128 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
48 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
64 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
192 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
72 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
96 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
128 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
48 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
64 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
32 CouplingC1 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
12 CouplingC2 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
16 CouplingC3 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
8 CouplingC1 MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
24 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]]^3 MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] - 8 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] + 4 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] -
72 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(2,0)}$[CTau[], CRho[]] - 12 CouplingC2 MetricA[CTau[], CRho[]]

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      MetricA^[illegible][CTau[], CRho[]] - 12 CouplingC2 MetricA[CTau[], CRho[]]
      ArealRadius^(1,0)[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]] +
    72 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius^(1,0)[CTau[], CRho[]]^2
      MetricA^(2,0)[CTau[], CRho[]] + 12 CouplingC2 MetricA[CTau[], CRho[]]^2
      ArealRadius^(1,0)[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]] -
    24 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius^(1,0)[CTau[], CRho[]]^2
      MetricA^(2,0)[CTau[], CRho[]] - 4 CouplingC2 MetricA[CTau[], CRho[]]^3
      ArealRadius^(1,0)[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]]^3 (8 FuncG4''[KineticXEScalar[]] ArealRadius^(0,1)[
        CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]]^4 MetricA^(0,1)[CTau[], CRho[]] -
    8 MetricA[CTau[], CRho[]] FuncG4''[KineticXEScalar[]] ArealRadius^(0,1)[
        CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]]^4 MetricA^(0,1)[CTau[], CRho[]] +
    112 CouplingC1 ArealRadius^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]]^3 +
    42 CouplingC2 ArealRadius^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]]^3 +
    56 CouplingC3 ArealRadius^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]]^3 +
    152 CouplingC1 MetricA^(0,1)[CTau[], CRho[]]^2 ArealRadius^(0,2)[CTau[], CRho[]] +
    57 CouplingC2 MetricA^(0,1)[CTau[], CRho[]]^2 ArealRadius^(0,2)[CTau[], CRho[]] +
    76 CouplingC3 MetricA^(0,1)[CTau[], CRho[]]^2 ArealRadius^(0,2)[CTau[], CRho[]] -
    152 CouplingC1 MetricA[CTau[], CRho[]]
      MetricA^(0,1)[CTau[], CRho[]]^2 ArealRadius^(0,2)[CTau[], CRho[]] -
    57 CouplingC2 MetricA[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]]^2
      ArealRadius^(0,2)[CTau[], CRho[]] - 76 CouplingC3 MetricA[CTau[], CRho[]]
      MetricA^(0,1)[CTau[], CRho[]]^2 ArealRadius^(0,2)[CTau[], CRho[]] +
    16 FuncG4''[KineticXEScalar[]] ArealRadius^(0,1)[CTau[], CRho[]]
      ClockProfile^(0,1)[CTau[], CRho[]]^3 ClockProfile^(0,2)[CTau[], CRho[]] -
    32 MetricA[CTau[], CRho[]] FuncG4''[KineticXEScalar[]]
      ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]]^3
      ClockProfile^(0,2)[CTau[], CRho[]] + 16 MetricA[CTau[], CRho[]]^2
      FuncG4''[KineticXEScalar[]] ArealRadius^(0,1)[CTau[], CRho[]]
      ClockProfile^(0,1)[CTau[], CRho[]]^3 ClockProfile^(0,2)[CTau[], CRho[]] +
    104 CouplingC1 ArealRadius^(0,1)[CTau[], CRho[]]
      MetricA^(0,1)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]] +
    39 CouplingC2 ArealRadius^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]]
      MetricA^(0,2)[CTau[], CRho[]] + 52 CouplingC3 ArealRadius^(0,1)[CTau[], CRho[]]
      MetricA^(0,1)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]] -
    104 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius^(0,1)[CTau[], CRho[]]
      MetricA^(0,1)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]] -
    39 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius^(0,1)[CTau[], CRho[]]
      MetricA^(0,1)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]] -
    52 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius^(0,1)[CTau[], CRho[]]
      MetricA^(0,1)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]] +
    64 CouplingC1 ArealRadius^(0,2)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]] +
    24 CouplingC2 ArealRadius^(0,2)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]] +
    32 CouplingC3 ArealRadius^(0,2)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]] -
    128 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]]
      MetricA^(0,2)[CTau[], CRho[]] - 48 CouplingC2 MetricA[CTau[], CRho[]]
      ArealRadius^(0,2)[CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]]
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ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] -
64 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(0,2)}$[CTau[], CRho[]] + 64 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] +
24 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(0,2)}$[CTau[], CRho[]] + 32 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] +
48 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] -
192 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,3)}$[CTau[], CRho[]] - 72 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] -
96 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,3)}$[CTau[], CRho[]] + 96 CouplingC1 MetricA[CTau[], CRho[]]^2
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,3)}$[CTau[], CRho[]] + 48 CouplingC3 MetricA[CTau[], CRho[]]^2
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] +
16 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] +
6 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] +
8 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,3)}$[CTau[], CRho[]] - 12 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,3)}$[CTau[], CRho[]] + 16 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,3)}$[CTau[], CRho[]] + 8 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] +
32 CouplingC1 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
12 CouplingC2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
16 CouplingC3 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
48 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
8 FuncG4$''$[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3 MetricA$^{(0,1)}$[CTau[],
CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]] FuncG4$''$[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3 MetricA$^{(0,1)}$[CTau[], CRho[]]

ClockProfile[CTau[], CRho[]] MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
8 MetricA[CTau[], CRho[]]^2 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
16 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
48 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
48 MetricA[CTau[], CRho[]]^2 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]]^3 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
40 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
15 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] - 20 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
40 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
15 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
20 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
8 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3

ClockProfile[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
AreaIRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{3}
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 MetricA[CTau[], CRho[]]^{2} FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{3}
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[
CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] -
24 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] +
24 MetricA[CTau[], CRho[]]^{2} FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] -
8 MetricA[CTau[], CRho[]]^{3} FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]] +
74 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
31 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} + 50 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
74 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
31 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
50 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
52 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
20 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
28 CouplingC3 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
104 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} - 40 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
56 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} + 52 CouplingC1 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
20 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} + 28 CouplingC3 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
34 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} -
10 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} -
6 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} +
68 CouplingC1 MetricA[CTau[], CRho[]]

ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 +
20 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^3 + 12 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 -
34 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^3 - 10 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 -
6 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^3 + 16 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] + 8 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
32 FuncG4$''$[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
96 MetricA[CTau[], CRho[]] FuncG4$''$[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
96 MetricA[CTau[], CRho[]]^2 FuncG4$''$[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
32 MetricA[CTau[], CRho[]]^3 FuncG4$''$[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
32 FuncG4$''$[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
128 MetricA[CTau[], CRho[]] FuncG4$''$[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
192 MetricA[CTau[], CRho[]]^2 FuncG4$''$[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 ClockProfile$^{(1,1)}$[CTau[], CRho[]] -

128 MetricA[CTau[], CRho[]]^{3} FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
32 MetricA[CTau[], CRho[]]^{4} FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
40 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
15 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] - 20 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
80 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
30 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
40 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
40 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
15 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
20 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
48 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
21 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] + 36 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
42 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
72 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
48 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
21 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
36 CouplingC3 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]

ArealRadius$^{(1,2)}$[CTau[], CRho[]] - 48 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] + 36 CouplingC2 MetricA[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
48 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] - 32 CouplingC1 MetricA[CTau[], CRho[]]^3
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^3 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] - 16 CouplingC3 MetricA[CTau[], CRho[]]^3
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
16 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
6 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
48 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
18 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] + 24 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] - 18 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
24 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] + 16 CouplingC1 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] + 8 CouplingC3 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
20 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
7 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
60 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
21 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 24 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
60 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 21 CouplingC2 MetricA[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
24 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 20 CouplingC1 MetricA[CTau[], CRho[]]^3
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
7 CouplingC2 MetricA[CTau[], CRho[]]^3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 8 CouplingC3 MetricA[CTau[], CRho[]]^3
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
2 FuncG4[KineticXEScalar[]] (-1 + MetricA[CTau[], CRho[]])^3

(ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) (-2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] + 2
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(2,0)}$[CTau[], CRho[]])) +
16 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
64 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
96 MetricA[CTau[], CRho[]]^{2} FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
64 MetricA[CTau[], CRho[]]^{3} FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
16 MetricA[CTau[], CRho[]]^{4} FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
16 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
80 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3}
ClockProfile$^{(2,0)}$[CTau[], CRho[]] + 160 MetricA[CTau[], CRho[]]^{2}
FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
160 MetricA[CTau[], CRho[]]^{3} FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3}
ClockProfile$^{(2,0)}$[CTau[], CRho[]] + 80 MetricA[CTau[], CRho[]]^{4}
FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]]^{5} FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[],
CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{3} ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
4 (-1 + MetricA[CTau[], CRho[]])^{2} FuncG4′[KineticXEScalar[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(4 ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) (-6 ClockProfile$^{(0,2)}$[CTau[], CRho[]] +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
6 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,0)}$[
CTau[], CRho[]])) - (-1 + MetricA[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} (2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(3 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] -
4 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,1)}$[
CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,0)}$[CTau[], CRho[]] (-6 ClockProfile$^{(0,2)}$[

ClockProfile$^{(1,0)}$[CTau[], CRho[]] (-6 ClockProfile$^{(0,2)}$[
CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 3 ArealRadius$^{(1,0)}$[CTau[],
CRho[]] (ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[
CTau[], CRho[]] + 2 (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) +
28 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 14 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
28 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
56 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
28 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
56 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
28 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
14 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
28 CouplingC3 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
40 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
16 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC3 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
120 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 48 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
72 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 120 CouplingC1 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
48 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 72 CouplingC3 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
40 CouplingC1 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 16 CouplingC2 MetricA[CTau[], CRho[]]^{3}
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
24 CouplingC3 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 60 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
17 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
180 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +

51 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
180 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
51 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
24 CouplingC3 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
60 CouplingC1 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
17 CouplingC2 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
24 CouplingC1 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 32 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
32 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 144 CouplingC1 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
48 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 48 CouplingC3 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 32 CouplingC2 MetricA[CTau[], CRho[]]^{3}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
32 CouplingC3 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 24 CouplingC1 MetricA[CTau[], CRho[]]^{4}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
8 CouplingC2 MetricA[CTau[], CRho[]]^{4} ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA[CTau[], CRho[]]^{4}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,1)}$[CTau[], CRho[]] - 48 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]

ArealRadius$^{(2,1)}$[CTau[], CRho[]] + 36 CouplingC2 MetricA[CTau[], CRho[]]^2
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
48 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,1)}$[CTau[], CRho[]] - 32 CouplingC1 MetricA[CTau[], CRho[]]^3
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^3 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,1)}$[CTau[], CRho[]] - 16 CouplingC3 MetricA[CTau[], CRho[]]^3
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
8 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
6 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
16 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -
24 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] - 18 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] + 24 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
18 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] + 48 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -
8 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] - 6 CouplingC2 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] + 64 CouplingC1 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
24 CouplingC2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
32 CouplingC3 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
256 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
96 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
128 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
384 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
144 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
192 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
256 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
96 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
128 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
64 CouplingC1 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
24 CouplingC2 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
32 CouplingC3 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
128 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] + 48 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
64 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] - 192 CouplingC1 MetricA[CTau[], CRho[]]^2

ArealRadius [CTau[], CRho[]] - 192 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
72 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] - 96 CouplingC3 MetricA[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
128 CouplingC1 MetricA[CTau[], CRho[]]^3 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] + 48 CouplingC2 MetricA[CTau[], CRho[]]^3
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
64 CouplingC3 MetricA[CTau[], CRho[]]^3 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] - 32 CouplingC1 MetricA[CTau[], CRho[]]^4
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^4 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] - 16 CouplingC3 MetricA[CTau[], CRho[]]^4
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
24 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
6 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] + 24 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
144 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 36 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] + 24 CouplingC2 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
24 CouplingC1 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 6 CouplingC2 MetricA[CTau[], CRho[]]^4
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
32 CouplingC1 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
12 CouplingC2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
16 CouplingC3 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
160 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
60 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
80 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
320 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
120 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
160 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
320 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
120 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
160 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
160 CouplingC1 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
60 CouplingC2 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
80 CouplingC3 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA[CTau[], CRho[]]^5 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^5 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]]^5 ArealRadius$^{(4,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^4 (-4 (-1 + MetricA[CTau[], CRho[]])

```
FuncG4''[KineticXEScalar[]] (ClockProfile^(0,1)[CTau[], CRho[]]^2 -
    (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(1,0)[CTau[], CRho[]]^2)
  (ClockProfile^(0,1)[CTau[], CRho[]]^2 MetricA^(1,0)[CTau[], CRho[]]^2 -
    2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(0,1)[CTau[], CRho[]]
     (2 MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
       MetricA^(0,1)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]]) +
    2 (-1 + MetricA[CTau[], CRho[]])^2 (2 ClockProfile^(1,1)[CTau[], CRho[]]^2 +
       (-2 ClockProfile^(0,2)[CTau[], CRho[]] +
          ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]])
        ClockProfile^(2,0)[CTau[], CRho[]])) +
 2 (-1 + MetricA[CTau[], CRho[]])^2 FuncG4'[KineticXEScalar[]]
  (-4 (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(0,1)[CTau[], CRho[]]
     (2 MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
       MetricA^(0,1)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]]) +
    ClockProfile^(0,1)[CTau[], CRho[]]^2 (3 MetricA^(1,0)[CTau[], CRho[]]^2 - 2
        (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,0)[CTau[], CRho[]]) +
    (-1 + MetricA[CTau[], CRho[]]) (4 (-1 + MetricA[CTau[], CRho[]])
        ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]
        ClockProfile^(2,0)[CTau[], CRho[]] + 8 (-1 + MetricA[CTau[], CRho[]])
        (ClockProfile^(1,1)[CTau[], CRho[]]^2 - ClockProfile^(0,2)[CTau[], CRho[]]
           ClockProfile^(2,0)[CTau[], CRho[]]) -
       ClockProfile^(1,0)[CTau[], CRho[]]^2 (MetricA^(1,0)[CTau[], CRho[]]^2 -
          2 (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,0)[CTau[], CRho[]]))) +
 (4 CouplingC1 + CouplingC2) (MetricA^(0,1)[CTau[], CRho[]]^2
     (7 MetricA^(1,0)[CTau[], CRho[]]^2 - 5 (-1 + MetricA[CTau[], CRho[]])
        MetricA^(2,0)[CTau[], CRho[]]) +
    (-1 + MetricA[CTau[], CRho[]]) MetricA^(0,1)[CTau[], CRho[]]
     (-9 MetricA^(1,0)[CTau[], CRho[]] MetricA^(1,1)[CTau[], CRho[]] + 5
        (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,1)[CTau[], CRho[]]) -
    (-1 + MetricA[CTau[], CRho[]]) (5 MetricA^(1,0)[CTau[], CRho[]]^4 - 12
        (-1 + MetricA[CTau[], CRho[]]) MetricA^(1,0)[CTau[], CRho[]]^2
        MetricA^(2,0)[CTau[], CRho[]] + 2 MetricA^(0,2)[CTau[], CRho[]]
        (MetricA^(1,0)[CTau[], CRho[]]^2 - (-1 + MetricA[CTau[], CRho[]])
           MetricA^(2,0)[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
        MetricA^(1,0)[CTau[], CRho[]] (-2 MetricA^(1,2)[CTau[], CRho[]] +
          5 (-1 + MetricA[CTau[], CRho[]]) MetricA^(3,0)[CTau[], CRho[]]) - 2
        (-1 + MetricA[CTau[], CRho[]]) (MetricA^(1,1)[CTau[], CRho[]]^2 -
          (-1 + MetricA[CTau[], CRho[]]) (2 MetricA^(2,0)[CTau[], CRho[]]^2 +
             MetricA^(2,2)[CTau[], CRho[]] - (-1 + MetricA[CTau[],
                CRho[]]) MetricA^(4,0)[CTau[], CRho[]])))))) /
  (4 ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^5)) == 0
```

```
SphericalEquations["EuclideanMetricEOM"][{"CPhi", "CPhi"}]
```

```
-((Sin[CTheta[]]^2 (8 (-1 + MetricA[CTau[], CRho[]])^3
       (1 - ArealRadius^(0,1)[CTau[], CRho[]]^2 - ArealRadius^(1,0)[CTau[], CRho[]]^2 +
         MetricA[CTau[], CRho[]] (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2))
```

(-2 CouplingC1 - CouplingC2 - 2 CouplingC3 -
2 (3 CouplingC1 + CouplingC2 + CouplingC3) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} -
6 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} -
2 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} -
2 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} + MetricA[CTau[], CRho[]]
(2 CouplingC1 + CouplingC2 + 2 CouplingC3 + 2 (3 CouplingC1 +
CouplingC2 + CouplingC3) ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2})) +
4 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^{2}
(6 (3 CouplingC1 + CouplingC2 + CouplingC3)
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{3} MetricA$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(MetricA$^{(0,1)}$[CTau[], CRho[]] (10 CouplingC1 + 3 CouplingC2 +
2 CouplingC3 - 2 (3 CouplingC1 + CouplingC2 + CouplingC3)
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}) + 16 (3 CouplingC1 +
CouplingC2 + CouplingC3) (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]) -
2 (3 CouplingC1 + CouplingC2 + CouplingC3) (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} (6 ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]])^{2} (2 ArealRadius$^{(0,2)}$[CTau[],
CRho[]] (-10 CouplingC1 - 3 CouplingC2 - 2 CouplingC3 +
2 (3 CouplingC1 + CouplingC2 + CouplingC3) ArealRadius$^{(1,0)}$[
CTau[], CRho[]]^{2}) + (10 CouplingC1 + 3 CouplingC2 +
2 CouplingC3) ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[
CTau[], CRho[]] - 2 (3 CouplingC1 + CouplingC2 + CouplingC3)
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{3} MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 (10 CouplingC1 + 3 CouplingC2 + 2 CouplingC3) (-1 + MetricA[CTau[],
CRho[]]) ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 12 (3 CouplingC1 +
CouplingC2 + CouplingC3) (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]])) +
ArealRadius[CTau[], CRho[]]^{2} (-1 + MetricA[CTau[], CRho[]])
(-8 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} -
3 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} -
4 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} -
32 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
12 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
16 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]

MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
32 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 -
12 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 -
16 CouplingC3 ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 +
64 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 +
24 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 +
32 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 -
32 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 -
12 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 -
16 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]^2 +
4 FuncG4[KineticXEScalar[]] (-1 + MetricA[CTau[], CRho[]])^3
(1 - ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 - ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 +
MetricA[CTau[], CRho[]] (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2)) +
8 (-1 + MetricA[CTau[], CRho[]])^2 FuncG4′[KineticXEScalar[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])^2 +
16 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
6 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
16 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] - 6 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 32 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
12 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 16 CouplingC3 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
64 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
24 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]

ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
4 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
8 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
4 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
28 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
2 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
8 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
28 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
2 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 8 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
4 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
4 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
8 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 2 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
8 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 4 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - 4 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
64 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
32 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
128 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
64 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
64 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
64 CouplingC1 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
32 CouplingC3 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
192 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 +
96 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 +
192 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
96 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
64 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 +
32 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,1)}$[CTau[], CRho[]]^2 -
64 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]

MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
12 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
128 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
64 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
128 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 24 CouplingC2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
384 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 72 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
384 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 72 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
128 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 24 CouplingC2 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
64 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
12 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
192 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
192 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
64 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
32 CouplingC1 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
12 CouplingC2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
16 CouplingC3 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
128 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
48 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 +
64 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
192 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2 -
72 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]^2

72 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]]^{2} -
96 CouplingC3 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]]^{2} +
128 CouplingC1 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(2,0)}$[CTau[], CRho[]]^{2} +
48 CouplingC2 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(2,0)}$[CTau[], CRho[]]^{2} +
64 CouplingC3 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(2,0)}$[CTau[], CRho[]]^{2} -
32 CouplingC1 MetricA[CTau[], CRho[]]^{4} ArealRadius$^{(2,0)}$[CTau[], CRho[]]^{2} -
12 CouplingC2 MetricA[CTau[], CRho[]]^{4} ArealRadius$^{(2,0)}$[CTau[], CRho[]]^{2} -
16 CouplingC3 MetricA[CTau[], CRho[]]^{4} ArealRadius$^{(2,0)}$[CTau[], CRho[]]^{2} -
8 CouplingC1 MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
24 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]]^{3} MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2}
MetricA$^{(2,0)}$[CTau[], CRho[]] - 8 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2}
MetricA$^{(2,0)}$[CTau[], CRho[]] + 4 CouplingC2 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] +
4 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] -
72 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}
MetricA$^{(2,0)}$[CTau[], CRho[]] - 12 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] +
72 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}
MetricA$^{(2,0)}$[CTau[], CRho[]] + 12 CouplingC2 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] -
24 CouplingC1 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}
MetricA$^{(2,0)}$[CTau[], CRho[]] - 4 CouplingC2 MetricA[CTau[], CRho[]]^{3}
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^{3} (-8 FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{4}
MetricA$^{(0,1)}$[CTau[], CRho[]] + 8 MetricA[CTau[], CRho[]]
FuncG4′′[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{4} MetricA$^{(0,1)}$[CTau[], CRho[]] -
112 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^{3} -
42 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^{3} -
56 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^{3} -
152 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
57 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
76 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
152 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
57 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(0,2)}$[CTau[], CRho[]] + 76 CouplingC3 MetricA[CTau[], CRho[]]

ArealRadius$^{(0,2)}$[CTau[], CRho[]] + 76 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
16 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3 ClockProfile$^{(0,2)}$[CTau[], CRho[]] +
32 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3
ClockProfile$^{(0,2)}$[CTau[], CRho[]] - 16 MetricA[CTau[], CRho[]]^2
FuncG4′′[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
104 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] -
39 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,2)}$[CTau[], CRho[]] - 52 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] +
104 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] +
39 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] +
52 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] -
64 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] -
24 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] -
32 CouplingC3 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] +
128 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(0,2)}$[CTau[], CRho[]] + 48 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] +
64 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(0,2)}$[CTau[], CRho[]] - 64 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] -
24 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(0,2)}$[CTau[], CRho[]] - 32 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] +
192 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,3)}$[CTau[], CRho[]] + 72 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] +
96 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,3)}$[CTau[], CRho[]] - 96 CouplingC1 MetricA[CTau[], CRho[]]^2
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,3)}$[CTau[], CRho[]] - 48 CouplingC3 MetricA[CTau[], CRho[]]^2
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,3)}$[CTau[], CRho[]] -
16 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] -
6 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] -
8 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] +

32 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,3)}$[CTau[], CRho[]] + 12 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,3)}$[CTau[], CRho[]] - 16 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,3)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,3)}$[CTau[], CRho[]] -
32 CouplingC1 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
12 CouplingC2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
16 CouplingC3 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
48 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
36 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,4)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,4)}$[CTau[], CRho[]] -
8 FuncG4″[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
16 MetricA[CTau[], CRho[]] FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
8 MetricA[CTau[], CRho[]]^2 FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
16 FuncG4″[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
48 MetricA[CTau[], CRho[]] FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
48 MetricA[CTau[], CRho[]]^2 FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
16 MetricA[CTau[], CRho[]]^3 FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
40 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
15 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 20 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]^2

ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
40 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
15 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
20 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 8 CouplingC3 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
16 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
6 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
8 FuncG4″[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
16 MetricA[CTau[], CRho[]] FuncG4″[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
8 MetricA[CTau[], CRho[]]^2 FuncG4″[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^3
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
8 FuncG4″[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]] +
24 MetricA[CTau[], CRho[]] FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]] -
24 MetricA[CTau[], CRho[]]^2 FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]] +
8 MetricA[CTau[], CRho[]]^3 FuncG4″[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]] -
74 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]

74 CouplingC1 ArealRadius [CTau[], CRho[]] MetricA [CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - 31 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
50 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
74 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
31 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
50 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
52 CouplingC1 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
20 CouplingC2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
28 CouplingC3 ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
104 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 + 40 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
56 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - 52 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
20 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - 28 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
34 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 +
10 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 +
6 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 -
68 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 -
20 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^3 - 12 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 +
34 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^3 + 10 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^3 +
6 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^3 - 16 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -

16 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
8 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
32 FuncG4$''$[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
96 MetricA[CTau[], CRho[]] FuncG4$''$[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
96 MetricA[CTau[], CRho[]]^{2} FuncG4$''$[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
32 MetricA[CTau[], CRho[]]^{3} FuncG4$''$[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
32 FuncG4$''$[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
128 MetricA[CTau[], CRho[]] FuncG4$''$[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
192 MetricA[CTau[], CRho[]]^{2} FuncG4$''$[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
128 MetricA[CTau[], CRho[]]^{3} FuncG4$''$[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
32 MetricA[CTau[], CRho[]]^{4} FuncG4$''$[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
40 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
15 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] + 20 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
80 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
30 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
40 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
40 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +

15 CouplingC2 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
20 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
48 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] - 21 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
36 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
42 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
72 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
21 CouplingC2 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
36 CouplingC3 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] + 48 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] - 36 CouplingC2 MetricA[CTau[], CRho[]]^{2}
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] + 32 CouplingC1 MetricA[CTau[], CRho[]]^{3}
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^{3} MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] + 16 CouplingC3 MetricA[CTau[], CRho[]]^{3}
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,2)}$[CTau[], CRho[]] +
16 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
6 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
8 CouplingC3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
48 CouplingC1 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
18 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] - 24 CouplingC3 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
48 CouplingC1 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]

48 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius [CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] + 18 CouplingC2 MetricA[CTau[], CRho[]]^{2}
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
24 CouplingC3 MetricA[CTau[], CRho[]]^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] - 16 CouplingC1 MetricA[CTau[], CRho[]]^{3}
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] -
6 CouplingC2 MetricA[CTau[], CRho[]]^{3} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,2)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA[CTau[], CRho[]]^{3}
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,2)}$[CTau[], CRho[]] +
20 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
7 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
60 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
21 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 24 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
60 CouplingC1 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] + 21 CouplingC2 MetricA[CTau[], CRho[]]^{2}
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
24 CouplingC3 MetricA[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 20 CouplingC1 MetricA[CTau[], CRho[]]^{3}
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
7 CouplingC2 MetricA[CTau[], CRho[]]^{3} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2}
ArealRadius$^{(2,0)}$[CTau[], CRho[]] - 8 CouplingC3 MetricA[CTau[], CRho[]]^{3}
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
2 FuncG4[KineticXEScalar[]] (-1 + MetricA[CTau[], CRho[]])^{3}
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) (-2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(2,0)}$[CTau[], CRho[]])) -
16 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
64 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
96 MetricA[CTau[], CRho[]]^{2} FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
64 MetricA[CTau[], CRho[]]^{3} FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
16 MetricA[CTau[], CRho[]]^{4} FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
16 FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]

16 FuncG4′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3 ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
80 MetricA[CTau[], CRho[]] FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3
ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
160 MetricA[CTau[], CRho[]]^2 FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3
ClockProfile$^{(2,0)}$[CTau[], CRho[]] + 160 MetricA[CTau[], CRho[]]^3
FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3 ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
80 MetricA[CTau[], CRho[]]^4 FuncG4′′[KineticXEScalar[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3
ClockProfile$^{(2,0)}$[CTau[], CRho[]] + 16 MetricA[CTau[], CRho[]]^5
FuncG4′′[KineticXEScalar[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^3 ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
4 (-1 + MetricA[CTau[], CRho[]])^2 FuncG4′[KineticXEScalar[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(4 ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]])
(-6 ClockProfile$^{(0,2)}$[CTau[], CRho[]] + ClockProfile$^{(1,0)}$[CTau[],
CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] + 6 (-1 + MetricA[
CTau[], CRho[]]) ClockProfile$^{(2,0)}$[CTau[], CRho[]])) -
(-1 + MetricA[CTau[], CRho[]]) (ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
(2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] - ArealRadius$^{(1,0)}$[
CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(3 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] -
4 (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,1)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(-6 ClockProfile$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[
CTau[], CRho[]] + 2 (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[
CTau[], CRho[]] + 3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[
CTau[], CRho[]] + 2 (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(2,0)}$[CTau[], CRho[]])))) -
28 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 14 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
28 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
56 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
28 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
56 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]

$$
\begin{aligned}
&\quad \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&28\, \text{CouplingC1}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\, \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&14\, \text{CouplingC2}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\, \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&28\, \text{CouplingC3}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\, \text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&40\, \text{CouplingC1}\, \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&16\, \text{CouplingC2}\, \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&24\, \text{CouplingC3}\, \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + \\
&120\, \text{CouplingC1}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]\, \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + 48\, \text{CouplingC2}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + \\
&72\, \text{CouplingC3}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]\, \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - 120\, \text{CouplingC1}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2 \\
&\quad \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&48\, \text{CouplingC2}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\, \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - 72\, \text{CouplingC3}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2 \\
&\quad \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + \\
&40\, \text{CouplingC1}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^3\, \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + 16\, \text{CouplingC2}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^3 \\
&\quad \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + \\
&24\, \text{CouplingC3}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^3\, \text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + 60\, \text{CouplingC1}\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + \\
&17\, \text{CouplingC2}\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + 8\, \text{CouplingC3}\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&180\, \text{CouplingC1}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&51\, \text{CouplingC2}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&24\, \text{CouplingC3}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + \\
&180\, \text{CouplingC1}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + \\
&51\, \text{CouplingC2}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] + \\
&24\, \text{CouplingC3}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^2\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&60\, \text{CouplingC1}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^3\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&17\, \text{CouplingC2}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^3\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] - \\
&8\, \text{CouplingC3}\, \text{MetricA}[\text{CTau}[], \text{CRho}[]]^3\, \text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]] \\
&\quad \text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\, \text{MetricA}^{(2,0)}[\text{CTau}[], \text{CRho}[]] +
\end{aligned}
$$

24 CouplingC1 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC2 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC3 ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 32 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 144 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
48 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 48 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 32 CouplingC2 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 24 CouplingC1 MetricA[CTau[], CRho[]]^4
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
8 CouplingC2 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 8 CouplingC3 MetricA[CTau[], CRho[]]^4
ArealRadius$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
32 CouplingC1 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
12 CouplingC2 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
16 CouplingC3 MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
96 CouplingC1 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
36 CouplingC2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,1)}$[CTau[], CRho[]] + 48 CouplingC3 MetricA[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,1)}$[CTau[], CRho[]] - 36 CouplingC2 MetricA[CTau[], CRho[]]^2
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
48 CouplingC3 MetricA[CTau[], CRho[]]^2 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,1)}$[CTau[], CRho[]] + 32 CouplingC1 MetricA[CTau[], CRho[]]^3
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^3 MetricA$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(2,1)}$[CTau[], CRho[]] + 16 CouplingC3 MetricA[CTau[], CRho[]]^3
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]] -
8 CouplingC1 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -
6 CouplingC2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -
16 CouplingC3 ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
24 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] + 18 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
48 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] - 24 CouplingC1 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] -

ArealRadius[CTau[], CRho[]] MetricA[CTau[], CRho[]]
18 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] - 48 CouplingC3 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
8 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] + 6 CouplingC2 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(2,1)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(2,1)}$[CTau[], CRho[]] - 64 CouplingC1 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
24 CouplingC2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
32 CouplingC3 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
256 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
96 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
128 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
384 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
144 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
192 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
256 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
96 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
128 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
64 CouplingC1 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
24 CouplingC2 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,2)}$[CTau[], CRho[]] -
32 CouplingC3 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(2,2)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
128 CouplingC1 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] - 48 CouplingC2 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
64 CouplingC3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] + 192 CouplingC1 MetricA[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
72 CouplingC2 MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] + 96 CouplingC3 MetricA[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
128 CouplingC1 MetricA[CTau[], CRho[]]^3 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] - 48 CouplingC2 MetricA[CTau[], CRho[]]^3
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] -
64 CouplingC3 MetricA[CTau[], CRho[]]^3 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] + 32 CouplingC1 MetricA[CTau[], CRho[]]^4
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^4 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(3,0)}$[CTau[], CRho[]] + 16 CouplingC3 MetricA[CTau[], CRho[]]^4
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] +
24 CouplingC1 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
6 CouplingC2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]

MetricA$^{(3,0)}$[CTau[], CRho[]] - 24 CouplingC2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
144 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] + 36 CouplingC2 MetricA[CTau[], CRho[]]^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
96 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] - 24 CouplingC2 MetricA[CTau[], CRho[]]^3
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] +
24 CouplingC1 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(3,0)}$[CTau[], CRho[]] + 6 CouplingC2 MetricA[CTau[], CRho[]]^4
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(3,0)}$[CTau[], CRho[]] -
32 CouplingC1 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
12 CouplingC2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
16 CouplingC3 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
160 CouplingC1 MetricA[CTau[], CRho[]] ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
60 CouplingC2 MetricA[CTau[], CRho[]] ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
80 CouplingC3 MetricA[CTau[], CRho[]] ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
320 CouplingC1 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
120 CouplingC2 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
160 CouplingC3 MetricA[CTau[], CRho[]]^2 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
320 CouplingC1 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
120 CouplingC2 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
160 CouplingC3 MetricA[CTau[], CRho[]]^3 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
160 CouplingC1 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
60 CouplingC2 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(4,0)}$[CTau[], CRho[]] -
80 CouplingC3 MetricA[CTau[], CRho[]]^4 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
32 CouplingC1 MetricA[CTau[], CRho[]]^5 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
12 CouplingC2 MetricA[CTau[], CRho[]]^5 ArealRadius$^{(4,0)}$[CTau[], CRho[]] +
16 CouplingC3 MetricA[CTau[], CRho[]]^5 ArealRadius$^{(4,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]]^4 (-4 (-1 + MetricA[CTau[], CRho[]])
FuncG4$''$[KineticXEScalar[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2)
(ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]) +
2 (-1 + MetricA[CTau[], CRho[]])^2 (2 ClockProfile$^{(1,1)}$[CTau[], CRho[]]^2 +
(-2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]])
ClockProfile$^{(2,0)}$[CTau[], CRho[]])) +
2 (-1 + MetricA[CTau[], CRho[]])^2 FuncG4$'$[KineticXEScalar[]]
(-4 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]) +
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2 (3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
2 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) +

(-1 + MetricA[CTau[], CRho[]]) (4 (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
8 (-1 + MetricA[CTau[], CRho[]]) (ClockProfile$^{(1,1)}$[CTau[], CRho[]]^2 -
ClockProfile$^{(0,2)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[
CTau[], CRho[]]) - ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2
(MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - 2 (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(2,0)}$[CTau[], CRho[]]))) +
(4 CouplingC1 + CouplingC2) (MetricA$^{(0,1)}$[CTau[], CRho[]]^2
(7 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - 5 (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(2,0)}$[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(0,1)}$[CTau[], CRho[]] (-9 MetricA$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] + 5 (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(2,1)}$[CTau[], CRho[]]) - (-1 + MetricA[CTau[], CRho[]])
(5 MetricA$^{(1,0)}$[CTau[], CRho[]]^4 - 12 (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(1,0)}$[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
2 MetricA$^{(0,2)}$[CTau[], CRho[]] (MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(1,0)}$[CTau[], CRho[]]
(-2 MetricA$^{(1,2)}$[CTau[], CRho[]] +
5 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(3,0)}$[CTau[],
CRho[]]) - 2 (-1 + MetricA[CTau[], CRho[]])
(MetricA$^{(1,1)}$[CTau[], CRho[]]^2 - (-1 + MetricA[CTau[], CRho[]])
(2 MetricA$^{(2,0)}$[CTau[], CRho[]]^2 + MetricA$^{(2,2)}$[CTau[],
CRho[]] - (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(4,0)}$[CTau[], CRho[]])))))) /
(4 ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^5)) == 0

EuclideanScalarEOM

```
SphericalEquations["EuclideanScalarEOM"][{}]
```

-((4 FuncK′[KineticXEScalar[]] (-((ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]) /
(-1 + MetricA[CTau[], CRho[]])) + ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])) / ArealRadius[CTau[], CRho[]]) -
(FuncK′[KineticXEScalar[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) /
(-1 + MetricA[CTau[], CRho[]])^2 + (2 FuncK′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) /
(-1 + MetricA[CTau[], CRho[]])^3 + (12 FuncG4′′[KineticXEScalar[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[],
CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]])^2

(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^4) -
(8 FuncG4$^{(3)}$[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[],
CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]])^2
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^5) -
(4 FuncK″[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])) /
(-1 + MetricA[CTau[], CRho[]])^2 + (16 FuncG4$^{(3)}$[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[],
CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]])^2
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])) /
(ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^4) -
(12 FuncG4″[KineticXEScalar[]] (ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])^2) /
(ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^4) +
(8 FuncG4$^{(3)}$[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])^2) /
(ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^5) +
(8 FuncG4$^{(3)}$[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2
(-ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])^2) /
(ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^4) -
(8 FuncG4″[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])

(2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(0,1)}$[CTau[], CRho[]] + ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(-2 (-1 + MetricA[CTau[], CRho[]])^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) +
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[
CTau[], CRho[]] (-ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) /
(ArealRadius[CTau[], CRho[]]^{3} (-1 + MetricA[CTau[], CRho[]])^{4}) -
(4 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))
(2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(0,1)}$[CTau[], CRho[]] + ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(-2 (-1 + MetricA[CTau[], CRho[]])^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) +
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[
CTau[], CRho[]] (-ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) /
(ArealRadius[CTau[], CRho[]]^{2} (-1 + MetricA[CTau[], CRho[]])^{5}) +
(8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])
(2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(0,1)}$[CTau[], CRho[]] + ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(-2 (-1 + MetricA[CTau[], CRho[]])^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) +
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[
CTau[], CRho[]] (-ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) /
(ArealRadius[CTau[], CRho[]]^{2} (-1 + MetricA[CTau[], CRho[]])^{4}) +
(16 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])
((-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} ClockProfile$^{(0,1)}$[
CTau[], CRho[]] + ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]

(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(0,2)}$[CTau[], CRho[]]) -
(-1 + MetricA[CTau[], CRho[]])^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]) + ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) (-ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] + (-1 + MetricA[CTau[], CRho[]])
(ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) /
(ArealRadius[CTau[], CRho[]]^{3} (-1 + MetricA[CTau[], CRho[]])^{4}) +
(8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))
((-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} ClockProfile$^{(0,1)}$[
CTau[], CRho[]] + ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(0,2)}$[CTau[], CRho[]]) -
(-1 + MetricA[CTau[], CRho[]])^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]) + ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) (-ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] + (-1 + MetricA[CTau[], CRho[]])
(ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) /
(ArealRadius[CTau[], CRho[]]^{2} (-1 + MetricA[CTau[], CRho[]])^{5}) +
(16 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(-ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])
((-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2}
ClockProfile$^{(0,1)}$[CTau[], CRho[]] +
ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(0,2)}$[CTau[], CRho[]]) -
(-1 + MetricA[CTau[], CRho[]])^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]) + ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) (-ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,2)}$[CTau[], CRho[]] + (-1 + MetricA[CTau[], CRho[]])
(ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) /
(ArealRadius[CTau[], CRho[]]^{2} (-1 + MetricA[CTau[], CRho[]])^{4}) -
2
FuncK′[
KineticXEScalar[]]
ClockProfile$^{(2,0)}$[
CTau[],
CRho[]] - 4 FuncK′′[
KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]^{2}

ClockProfile$^{(2,0)}$[
CTau[],
CRho[]] +
(24 FuncG4′′[KineticXEScalar[]] ($\frac{\text{ArealRadius}^{(0,1)}\text{[CTau[],CRho[]] ClockProfile}^{(0,1)}\text{[CTau[],CRho[]]}}{\text{-1+MetricA[CTau[],CRho[]]}}$ -
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]])^2
ClockProfile$^{(2,0)}$[CTau[], CRho[]]) / ArealRadius[CTau[], CRho[]]^2 +
(16 FuncG4$^{(3)}$[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2
($\frac{\text{ArealRadius}^{(0,1)}\text{[CTau[],CRho[]] ClockProfile}^{(0,1)}\text{[CTau[],CRho[]]}}{\text{-1+MetricA[CTau[],CRho[]]}}$ -
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]])^2
ClockProfile$^{(2,0)}$[CTau[], CRho[]]) / ArealRadius[CTau[], CRho[]]^2 -
(24 FuncG4′′[KineticXEScalar[]] (ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))
ClockProfile$^{(2,0)}$[CTau[], CRho[]]) / (ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]])^3) +
(16 FuncG4$^{(3)}$[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]^2
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))
ClockProfile$^{(2,0)}$[CTau[], CRho[]]) / (ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]])^4) +
(16 FuncG4$^{(3)}$[KineticXEScalar[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]^2
(-ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))
ClockProfile$^{(2,0)}$[CTau[], CRho[]]) / (ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]])^3) +
(8 FuncG4′′[KineticXEScalar[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2
ClockProfile$^{(0,1)}$[CTau[], CRho[]] + ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(-2 (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) +

$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])\,\text{ArealRadius}^{(1,0)}[$$
$$\text{CTau}[], \text{CRho}[]]\,(-\text{ClockProfile}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]]+$$
$$2\,(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])\,\text{ClockProfile}^{(1,1)}[\text{CTau}[], \text{CRho}[]]))$$
$$\text{ClockProfile}^{(2,0)}[\text{CTau}[], \text{CRho}[]])/(\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]^2$$
$$(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])^3)+$$
$$(16\,\text{FuncG4}''[\text{KineticXEScalar}[]]\,\text{ClockProfile}^{(0,1)}[\text{CTau}[], \text{CRho}[]]$$
$$(\text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,(\frac{\text{ArealRadius}^{(0,1)}[\text{CTau}[],\text{CRho}[]]\,\text{ClockProfile}^{(0,1)}[\text{CTau}[],\text{CRho}[]]}{-1+\text{MetricA}[\text{CTau}[],\text{CRho}[]]}-$$
$$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\,\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]])+$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,((\text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,\text{ClockProfile}^{(0,1)}[\text{CTau}[],$$
$$\text{CRho}[]]\,\text{MetricA}^{(0,1)}[\text{CTau}[], \text{CRho}[]])/(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])^2+$$
$$(\text{ClockProfile}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,\text{ArealRadius}^{(0,2)}[\text{CTau}[], \text{CRho}[]]+$$
$$\text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,\text{ClockProfile}^{(0,2)}[\text{CTau}[], \text{CRho}[]])/$$
$$(1-\text{MetricA}[\text{CTau}[], \text{CRho}[]])+\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$$
$$\text{ArealRadius}^{(1,1)}[\text{CTau}[], \text{CRho}[]]+$$
$$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\,\text{ClockProfile}^{(1,1)}[\text{CTau}[], \text{CRho}[]]))$$
$$\text{ClockProfile}^{(2,0)}[\text{CTau}[], \text{CRho}[]])/(\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]^2$$
$$(1-\text{MetricA}[\text{CTau}[], \text{CRho}[]]))+$$
$$(16\,\text{FuncG4}''[\text{KineticXEScalar}[]]\,\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$$
$$(-\frac{\text{ArealRadius}^{(0,1)}[\text{CTau}[],\text{CRho}[]]\,\text{ClockProfile}^{(0,1)}[\text{CTau}[],\text{CRho}[]]}{-1+\text{MetricA}[\text{CTau}[],\text{CRho}[]]}+$$
$$\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\,\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]])$$
$$(-\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2\,\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]]+$$
$$(\text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,(\text{ClockProfile}^{(0,1)}[\text{CTau}[], \text{CRho}[]]$$
$$((-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])\,\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]+$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]])-$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])$$
$$\text{ClockProfile}^{(1,1)}[\text{CTau}[], \text{CRho}[]]))/(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])^2+$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,(-((\text{ClockProfile}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,\text{ArealRadius}^{(1,1)}[$$
$$\text{CTau}[], \text{CRho}[]])/(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]]))+$$
$$\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\,\text{ArealRadius}^{(2,0)}[\text{CTau}[], \text{CRho}[]])+$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$$
$$\text{ClockProfile}^{(2,0)}[\text{CTau}[], \text{CRho}[]]))/\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]^3+$$
$$(16\,\text{FuncG4}''[\text{KineticXEScalar}[]]\,\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$$
$$(\frac{\text{ClockProfile}^{(0,1)}[\text{CTau}[],\text{CRho}[]]\,\text{MetricA}^{(0,1)}[\text{CTau}[],\text{CRho}[]]}{2-2\,\text{MetricA}[\text{CTau}[],\text{CRho}[]]}+\text{ClockProfile}^{(0,2)}[\text{CTau}[], \text{CRho}[]]-$$
$$\frac{1}{2}\,\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\,\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]])$$
$$(-\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]^2\,\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]]+$$
$$(\text{ArealRadius}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,(\text{ClockProfile}^{(0,1)}[\text{CTau}[], \text{CRho}[]]$$
$$((-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])\,\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]+$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,\text{MetricA}^{(1,0)}[\text{CTau}[], \text{CRho}[]])-$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])$$
$$\text{ClockProfile}^{(1,1)}[\text{CTau}[], \text{CRho}[]]))/(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]])^2+$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,(-((\text{ClockProfile}^{(0,1)}[\text{CTau}[], \text{CRho}[]]\,\text{ArealRadius}^{(1,1)}[$$
$$\text{CTau}[], \text{CRho}[]])/(-1+\text{MetricA}[\text{CTau}[], \text{CRho}[]]))+$$
$$\text{ClockProfile}^{(1,0)}[\text{CTau}[], \text{CRho}[]]\,\text{ArealRadius}^{(2,0)}[\text{CTau}[], \text{CRho}[]])+$$
$$\text{ArealRadius}[\text{CTau}[], \text{CRho}[]]\,\text{ArealRadius}^{(1,0)}[\text{CTau}[], \text{CRho}[]]$$

ClockProfile$^{(2,0)}$[CTau[], CRho[]])) /
(ArealRadius[CTau[], CRho[]]^2 (1 - MetricA[CTau[], CRho[]])) -
(32 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]
($\frac{\text{ClockProfile}^{(0,1)}\text{[CTau[],CRho[]] MetricA}^{(1,0)}\text{[CTau[],CRho[]]}}{\text{2-2 MetricA[CTau[],CRho[]]}}$ + ClockProfile$^{(1,1)}$[CTau[], CRho[]])
(-ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
((-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,1)}$[CTau[], CRho[]])) / (-1 + MetricA[CTau[], CRho[]])^2 +
ArealRadius[CTau[], CRho[]] (-((ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[
CTau[], CRho[]]) / (-1 + MetricA[CTau[], CRho[]])) +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]])) /
(ArealRadius[CTau[], CRho[]]^2 (1 - MetricA[CTau[], CRho[]])) -
(16 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]]
(-ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
((-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,1)}$[CTau[], CRho[]])) / (-1 + MetricA[CTau[], CRho[]])^2 +
ArealRadius[CTau[], CRho[]] (-((ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[
CTau[], CRho[]]) / (-1 + MetricA[CTau[], CRho[]])) +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]])) /
ArealRadius[CTau[], CRho[]]^2 - (4 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) -
2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,1)}$[CTau[], CRho[]]) -
2 (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
ArealRadius[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^4) +
(8 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]

ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) -
2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,1)}$[CTau[], CRho[]]) -
2 (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
ArealRadius[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^4) -
(16 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(- $\frac{\text{ArealRadius}^{(0,1)}\text{[CTau[],CRho[]] ClockProfile}^{(0,1)}\text{[CTau[],CRho[]]}}{\text{-1+MetricA[CTau[],CRho[]]}}$ +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]])
((ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) -
2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,1)}$[CTau[], CRho[]])) /
(2 (-1 + MetricA[CTau[], CRho[]])^2) + ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(-ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
ArealRadius[CTau[], CRho[]]^3 + (16 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]]
((ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]) - 2 ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])) /
(2 (-1 + MetricA[CTau[], CRho[]])^2) + ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(-ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
ArealRadius[CTau[], CRho[]]^2 + (8 FuncG4′[KineticXEScalar[]]
(- $\frac{\text{ArealRadius}^{(0,1)}\text{[CTau[],CRho[]]}^3\text{ ClockProfile}^{(0,1)}\text{[CTau[],CRho[]]}}{\text{(-1+MetricA[CTau[],CRho[]])}^2}$ + (ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2
(-ArealRadius[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] + 2 (-1 + MetricA[CTau[], CRho[]])^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
(2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] - ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]))) / (2 (-1 + MetricA[CTau[], CRho[]])^3) +
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] ((-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]

MetricA$^{(1,0)}$[CTau[], CRho[]]) - 2 ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])) /
(-1 + MetricA[CTau[], CRho[]])^2 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
(-ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
ArealRadius[CTau[], CRho[]]^3 - (8 FuncG4′[KineticXEScalar[]]
(- $\frac{\text{ArealRadius}^{(0,1)}\text{[CTau[],CRho[]]}^3\ \text{ClockProfile}^{(0,1)}\text{[CTau[],CRho[]]}}{\text{(-1+MetricA[CTau[],CRho[]])}^2}$ +
(ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 (-ArealRadius[CTau[], CRho[]] ClockProfile$^{(0,1)}$[
CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] + ArealRadius[CTau[],
CRho[]] (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(0,2)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]])^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])) / (-1 + MetricA[CTau[], CRho[]])^3 +
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ((-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(0,1)}$[CTau[],
CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 + ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) -
(-1 + MetricA[CTau[], CRho[]]) (ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] + 2 ArealRadius$^{(1,0)}$[
CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) /
(-1 + MetricA[CTau[], CRho[]])^2 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(-ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] (-((ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[
CTau[], CRho[]]) / (-1 + MetricA[CTau[], CRho[]])) +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
ArealRadius[CTau[], CRho[]]^3 - (2 FuncG4′[KineticXEScalar[]]
(-2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^2
ArealRadius$^{(0,2)}$[CTau[], CRho[]] (ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]]) +
4 (1 - MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]
((1 - MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]) - ArealRadius[CTau[], CRho[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(0,2)}$[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
(-ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) (ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] + ArealRadius$^{(1,0)}$[
CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) -
ArealRadius[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
((1 - MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]

(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]) - ArealRadius[CTau[], CRho[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(0,2)}$[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
(-ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) (ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] + ArealRadius$^{(1,0)}$[
CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]])))) +
2 ArealRadius[CTau[], CRho[]]^2 (2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,2)}$[CTau[], CRho[]]) +
(1 - MetricA[CTau[], CRho[]]) (ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[],
CRho[]] ClockProfile$^{(0,2)}$[CTau[], CRho[]] + ClockProfile$^{(0,1)}$[CTau[],
CRho[]] (MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,2)}$[CTau[], CRho[]])) +
(-1 + MetricA[CTau[], CRho[]])^2 (2 ArealRadius$^{(0,2)}$[CTau[], CRho[]]
ClockProfile$^{(0,2)}$[CTau[], CRho[]] + ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(0,3)}$[CTau[], CRho[]] + ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,3)}$[CTau[], CRho[]]) - 2 (-1 + MetricA[CTau[], CRho[]])^3
ArealRadius$^{(1,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]])^3 ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,2)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])^3
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,2)}$[CTau[], CRho[]]) +
ArealRadius[CTau[], CRho[]] (1 - MetricA[CTau[], CRho[]]) MetricA$^{(1,0)}$[CTau[], CRho[]]
((-1 + MetricA[CTau[], CRho[]])^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]] - ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] ((-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]) - ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]] (1 - MetricA[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]]^3 (-1 + MetricA[CTau[], CRho[]])^4) -
(2 FuncG4′[KineticXEScalar[]]
(4 (1 - MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]
((1 - MetricA[CTau[], CRho[]]) ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]) - ArealRadius[CTau[], CRho[]]

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ClockProfile^(1,0)[CTau[], CRho[]]) - ArealRadius[CTau[], CRho[]]
    (ArealRadius^(0,1)[CTau[], CRho[]] (ClockProfile^(0,1)[CTau[], CRho[]]
            MetricA^(0,1)[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
            ClockProfile^(0,2)[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
        (-ClockProfile^(0,1)[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] +
          (-1 + MetricA[CTau[], CRho[]]) (ClockProfile^(1,0)[CTau[], CRho[]]
              ArealRadius^(1,1)[CTau[], CRho[]] + ArealRadius^(1,0)[
               CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]])))) -
ArealRadius[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]]
  ((1 - MetricA[CTau[], CRho[]]) ArealRadius^(0,1)[CTau[], CRho[]]
      (ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]] -
        (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(1,0)[CTau[], CRho[]]
         ClockProfile^(1,0)[CTau[], CRho[]]) - ArealRadius[CTau[], CRho[]]
      (ArealRadius^(0,1)[CTau[], CRho[]] (ClockProfile^(0,1)[CTau[], CRho[]]
            MetricA^(0,1)[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
            ClockProfile^(0,2)[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
        (-ClockProfile^(0,1)[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] +
          (-1 + MetricA[CTau[], CRho[]]) (ClockProfile^(1,0)[CTau[], CRho[]]
              ArealRadius^(1,1)[CTau[], CRho[]] + ArealRadius^(1,0)[
               CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]])))) -
2 ArealRadius[CTau[], CRho[]] ((-1 + MetricA[CTau[], CRho[]])^2
      ArealRadius^(0,2)[CTau[], CRho[]] (ArealRadius^(0,1)[CTau[], CRho[]]
         ClockProfile^(0,1)[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
         ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,0)[CTau[], CRho[]]) -
    ArealRadius[CTau[], CRho[]] (2 ArealRadius^(0,1)[CTau[], CRho[]]
         ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]]^2 -
        (-1 + MetricA[CTau[], CRho[]]) MetricA^(0,1)[CTau[], CRho[]]
         (ClockProfile^(0,1)[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] +
           ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,2)[CTau[], CRho[]]) +
        (1 - MetricA[CTau[], CRho[]]) (ArealRadius^(0,1)[CTau[], CRho[]]
            MetricA^(0,1)[CTau[], CRho[]] ClockProfile^(0,2)[CTau[], CRho[]] +
           ClockProfile^(0,1)[CTau[], CRho[]] (MetricA^(0,1)[CTau[], CRho[]]
               ArealRadius^(0,2)[CTau[], CRho[]] + ArealRadius^(0,1)[
                CTau[], CRho[]] MetricA^(0,2)[CTau[], CRho[]])) +
        (-1 + MetricA[CTau[], CRho[]])^2 (2 ArealRadius^(0,2)[CTau[], CRho[]]
            ClockProfile^(0,2)[CTau[], CRho[]] + ClockProfile^(0,1)[CTau[], CRho[]]
            ArealRadius^(0,3)[CTau[], CRho[]] + ArealRadius^(0,1)[CTau[], CRho[]]
            ClockProfile^(0,3)[CTau[], CRho[]]) - 2 (-1 + MetricA[CTau[], CRho[]])^3
         ArealRadius^(1,1)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
        (-1 + MetricA[CTau[], CRho[]])^3 ClockProfile^(1,0)[CTau[], CRho[]]
         ArealRadius^(1,2)[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])^3
         ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,2)[CTau[], CRho[]])) +
ArealRadius[CTau[], CRho[]] (1 - MetricA[CTau[], CRho[]]) MetricA^(1,0)[CTau[], CRho[]]
  ((-1 + MetricA[CTau[], CRho[]])^2 ArealRadius^(1,0)[CTau[], CRho[]]^2
      ClockProfile^(1,0)[CTau[], CRho[]] - ArealRadius^(0,1)[CTau[], CRho[]]
      (ClockProfile^(0,1)[CTau[], CRho[]] ((-1 + MetricA[CTau[], CRho[]])
            ArealRadius^(1,0)[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]
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MetricA$^{(1,0)}$[CTau[], CRho[]]) - ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]] (1 - MetricA[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^{2}
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]]^{3} (-1 + MetricA[CTau[], CRho[]])^{4}) +
(4
FuncG4′′[
KineticXEScalar[]] ClockProfile$^{(1,0)}$[
CTau[],
CRho[]]
(-ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])
(2 ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[],
CRho[]] - (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(1,1)}$[CTau[], CRho[]]) -
(-1 + MetricA[CTau[], CRho[]]) (4 ClockProfile$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[],
CRho[]] - ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
2 MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
4 ClockProfile$^{(1,2)}$[CTau[], CRho[]] -
4 MetricA[CTau[], CRho[]] ClockProfile$^{(1,2)}$[CTau[], CRho[]] -
2 MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] + 2 MetricA[CTau[],
CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^{4}) -
(FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[
CTau[],
CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])
(2 ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(1,1)}$[CTau[], CRho[]]) -
(-1 + MetricA[CTau[], CRho[]]) (4 ClockProfile$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[],
CRho[]] - ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
2 MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
4 ClockProfile$^{(1,2)}$[CTau[], CRho[]] -
4 MetricA[CTau[], CRho[]] ClockProfile$^{(1,2)}$[CTau[], CRho[]] -
2 MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] + 2 MetricA[CTau[],
CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) /
(-1 + MetricA[CTau[], CRho[]])^{5} - (2 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[
CTau[],
CRho[]] ClockProfile$^{(2,0)}$[

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            CTau[],
            CRho[]]
          (2 ClockProfile^(0,1)[CTau[], CRho[]]
                (2 MetricA^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
                    (-1 + MetricA[CTau[], CRho[]]) MetricA^(1,1)[CTau[], CRho[]]) -
              (-1 + MetricA[CTau[], CRho[]]) (4 ClockProfile^(0,2)[CTau[], CRho[]] MetricA^(1,0)[CTau[],
                        CRho[]] - ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]^2 +
                    2 MetricA^(0,1)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] +
                    4 ClockProfile^(1,2)[CTau[], CRho[]] -
                    4 MetricA[CTau[], CRho[]] ClockProfile^(1,2)[CTau[], CRho[]] -
                    2 MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]] + 2 MetricA[CTau[],
                        CRho[]] MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]]))) /
      (-1 + MetricA[CTau[], CRho[]])^3 + (2 FuncG4′[KineticXEScalar[]]
          (-4 ArealRadius^(0,1)[CTau[], CRho[]]
                ((ArealRadius^(0,1)[CTau[], CRho[]]^2 ClockProfile^(0,1)[CTau[], CRho[]]) /
                      (-1 + MetricA[CTau[], CRho[]]) +
                    1/2 ArealRadius^(0,1)[CTau[], CRho[]] (-2 ArealRadius^(1,0)[CTau[], CRho[]]
                            ClockProfile^(1,0)[CTau[], CRho[]] + (ArealRadius[CTau[], CRho[]]
                                (ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]] -
                                    (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile^(0,2)[CTau[],
                                            CRho[]] - ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,0)[
                                            CTau[], CRho[]]))) / (-1 + MetricA[CTau[], CRho[]])^2) +
                    ArealRadius[CTau[], CRho[]] ArealRadius^(1,0)[CTau[], CRho[]]
                      ((ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]) /
                            (2 - 2 MetricA[CTau[], CRho[]]) + ClockProfile^(1,1)[CTau[], CRho[]])) -
              4 ArealRadius^(0,1)[CTau[], CRho[]] (ArealRadius^(0,1)[CTau[], CRho[]]
                      ((ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]]) /
                            (-1 + MetricA[CTau[], CRho[]]) - ArealRadius^(1,0)[CTau[], CRho[]]
                            ClockProfile^(1,0)[CTau[], CRho[]]) + ArealRadius[CTau[], CRho[]]
                      ((ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]]
                                MetricA^(0,1)[CTau[], CRho[]]) / (-1 + MetricA[CTau[], CRho[]])^2 +
                          (ClockProfile^(0,1)[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] +
                                ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,2)[CTau[], CRho[]]) /
                            (1 - MetricA[CTau[], CRho[]]) + ClockProfile^(1,0)[CTau[], CRho[]]
                            ArealRadius^(1,1)[CTau[], CRho[]] +
                          ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]])) -
              (2 ArealRadius[CTau[], CRho[]]^2 ArealRadius^(0,1)[CTau[], CRho[]]
                    (ClockProfile^(0,1)[CTau[], CRho[]] (2 MetricA^(0,1)[CTau[], CRho[]]^2 -
                              (-1 + MetricA[CTau[], CRho[]]) (MetricA^(0,2)[CTau[], CRho[]] -
                                    MetricA^(1,0)[CTau[], CRho[]]^2)) + (-1 + MetricA[CTau[], CRho[]])
                          (MetricA^(0,1)[CTau[], CRho[]] (-3 ClockProfile^(0,2)[CTau[], CRho[]] +
                                    ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]) +
                              (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile^(0,3)[CTau[], CRho[]] -
                                    3 MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
                                    ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,1)[CTau[], CRho[]])))) /
                (-1 + MetricA[CTau[], CRho[]])^3 + (ArealRadius[CTau[], CRho[]]^2
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```
ArealRadius^(1,0)[CTau[], CRho[]] (2 ClockProfile^(0,1)[CTau[], CRho[]]
      (2 MetricA^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
        (-1 + MetricA[CTau[], CRho[]]) MetricA^(1,1)[CTau[], CRho[]]) -
     (-1 + MetricA[CTau[], CRho[]]) (4 ClockProfile^(0,2)[CTau[], CRho[]]
         MetricA^(1,0)[CTau[], CRho[]] - ClockProfile^(1,0)[CTau[], CRho[]]
         MetricA^(1,0)[CTau[], CRho[]]^2 + 2 MetricA^(0,1)[CTau[], CRho[]]
         ClockProfile^(1,1)[CTau[], CRho[]] + 4 ClockProfile^(1,2)[CTau[], CRho[]] -
        4 MetricA[CTau[], CRho[]] ClockProfile^(1,2)[CTau[], CRho[]] - 2
         MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]] + 2
         MetricA[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]
         ClockProfile^(2,0)[CTau[], CRho[]]))) / (-1 + MetricA[CTau[], CRho[]])^2)) /
 (ArealRadius[CTau[], CRho[]]^3 (1 - MetricA[CTau[], CRho[]])) +
(2
   FuncG4′[
    KineticXEScalar[]]
   ((4 ArealRadius^(0,1)[CTau[], CRho[]]^2 (-((ArealRadius^(0,1)[CTau[], CRho[]]
                ClockProfile^(0,1)[CTau[], CRho[]]) / (-1 + MetricA[CTau[], CRho[]])) +
           ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,0)[CTau[], CRho[]])) /
      ArealRadius[CTau[], CRho[]] + ArealRadius^(1,0)[CTau[], CRho[]]
      ClockProfile^(1,0)[CTau[], CRho[]]
      (-2 ArealRadius^(0,2)[CTau[], CRho[]] + ArealRadius^(1,0)[CTau[], CRho[]]
         MetricA^(1,0)[CTau[], CRho[]]) + (ArealRadius^(0,1)[CTau[], CRho[]]^2
        (-4 ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]] +
          (-1 + MetricA[CTau[], CRho[]]) (4 ClockProfile^(0,2)[CTau[], CRho[]] -
             ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]))) /
      (-1 + MetricA[CTau[], CRho[]])^2 + (ArealRadius^(0,1)[CTau[], CRho[]]
        (4 ClockProfile^(0,1)[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] +
          MetricA^(0,1)[CTau[], CRho[]] ArealRadius^(1,0)[CTau[], CRho[]]
           ClockProfile^(1,0)[CTau[], CRho[]] - 2 (-1 + MetricA[CTau[], CRho[]])
           (ClockProfile^(1,0)[CTau[], CRho[]] ArealRadius^(1,1)[CTau[], CRho[]] + 2
              ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]]))) /
      (-1 + MetricA[CTau[], CRho[]]) - (ArealRadius[CTau[], CRho[]] (ArealRadius^(0,1)[CTau[],
            CRho[]] (ClockProfile^(0,1)[CTau[], CRho[]] (5 MetricA^(0,1)[CTau[], CRho[]]^2 -
                (-1 + MetricA[CTau[], CRho[]]) (2 MetricA^(0,2)[CTau[], CRho[]] -
                   MetricA^(1,0)[CTau[], CRho[]]^2)) +
             (-1 + MetricA[CTau[], CRho[]]) (MetricA^(0,1)[CTau[], CRho[]]
                 (-8 ClockProfile^(0,2)[CTau[], CRho[]] + 3 ClockProfile^(1,0)[
                     CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]) +
                2 (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile^(0,3)[
                     CTau[], CRho[]] - 2 MetricA^(1,0)[CTau[], CRho[]]
                    ClockProfile^(1,1)[CTau[], CRho[]] - ClockProfile^(1,0)[
                     CTau[], CRho[]] MetricA^(1,1)[CTau[], CRho[]]))) -
          (-1 + MetricA[CTau[], CRho[]]) (ClockProfile^(0,1)[CTau[], CRho[]]
              (MetricA^(0,1)[CTau[], CRho[]] (2 ArealRadius^(0,2)[CTau[], CRho[]] +
                   3 ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[],
                     CRho[]]) - 2 (-1 + MetricA[CTau[], CRho[]])
                  (MetricA^(1,0)[CTau[], CRho[]] ArealRadius^(1,1)[CTau[], CRho[]]
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(MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[],
CRho[]])) - 2 (-1 + MetricA[CTau[], CRho[]])
(ClockProfile$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + ArealRadius$^{(0,2)}$[CTau[], CRho[]]
(2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] - ClockProfile$^{(1,0)}$[
CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] + 2 ArealRadius$^{(1,1)}$[CTau[],
CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] - 2 MetricA[CTau[],
CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[
CTau[], CRho[]] + 2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,2)}$[CTau[], CRho[]] - 2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,2)}$[CTau[],
CRho[]] - ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[],
CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] + MetricA[CTau[],
CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[
CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]])))) /
(2 (-1 + MetricA[CTau[], CRho[]])^{3}))) / (ArealRadius[CTau[], CRho[]]^{2}
(1 - MetricA[CTau[], CRho[]])) + (2
FuncG4′[
KineticXEScalar[]]
((4 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} (-((ArealRadius$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(0,1)}$[CTau[], CRho[]]) / (-1 + MetricA[CTau[], CRho[]])) +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]])) /
ArealRadius[CTau[], CRho[]] + ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(-2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] + ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]) + (ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2}
(-4 ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) (4 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))) /
(-1 + MetricA[CTau[], CRho[]])^{2} + (ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(4 ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] - 2 (-1 + MetricA[CTau[], CRho[]])
(ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] + 2
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]]))) /
(-1 + MetricA[CTau[], CRho[]]) + (ArealRadius[CTau[], CRho[]]
(-ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (5 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} -
(-1 + MetricA[CTau[], CRho[]]) (2 MetricA$^{(0,2)}$[CTau[], CRho[]] -
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2})) +
(-1 + MetricA[CTau[], CRho[]]) (MetricA$^{(0,1)}$[CTau[], CRho[]]
(-8 ClockProfile$^{(0,2)}$[CTau[], CRho[]] + 3 ClockProfile$^{(1,0)}$[
CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
2 (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,3)}$[

CTau[], CRho[]] - 2 MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] - ClockProfile$^{(1,0)}$[
CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]]))) +
(-1 + MetricA[CTau[], CRho[]]) (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(MetricA$^{(0,1)}$[CTau[], CRho[]] (2 ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
3 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[],
CRho[]]) - 2 (-1 + MetricA[CTau[], CRho[]])
(MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[],
CRho[]])) - 2 (-1 + MetricA[CTau[], CRho[]])
(ClockProfile$^{(0,2)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + ArealRadius$^{(0,2)}$[CTau[], CRho[]]
(2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] - ClockProfile$^{(1,0)}$[
CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
MetricA$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] + 2 ArealRadius$^{(1,1)}$[CTau[],
CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] - 2 MetricA[CTau[],
CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[
CTau[], CRho[]] + 2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,2)}$[CTau[], CRho[]] - 2 MetricA[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,2)}$[CTau[],
CRho[]] - ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[],
CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] + MetricA[CTau[],
CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[
CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]])))) /
(2 (-1 + MetricA[CTau[], CRho[]])^{3}))) / (ArealRadius[CTau[], CRho[]]^{2}
(1 - MetricA[CTau[], CRho[]])) + (FuncG4′[
KineticXEScalar[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]])
(2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))
(-4 + 4 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^{2} + 4 ArealRadius[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] +
8 ArealRadius[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
4 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} -
4 ArealRadius[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] -
ArealRadius[CTau[], CRho[]]^{2} MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
8 ArealRadius[CTau[], CRho[]]
ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
4 MetricA[CTau[], CRho[]]^{2} (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} +
2 ArealRadius[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]) -
2 ArealRadius[CTau[], CRho[]]^{2} MetricA$^{(2,0)}$[CTau[], CRho[]] -
2 MetricA[CTau[], CRho[]]

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              2 MetricA[CTau[], CRho[]]
                (2 ArealRadius^(0,1)[CTau[], CRho[]]^2 +
                    4 (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2) + ArealRadius[CTau[], CRho[]]
                      (4 ArealRadius^(0,2)[CTau[], CRho[]] - 2 ArealRadius^(1,0)[CTau[], CRho[]]
                            MetricA^(1,0)[CTau[], CRho[]] + 8 ArealRadius^(2,0)[CTau[], CRho[]]) -
                    ArealRadius[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]]))) /
  (2 ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^4) -
(FuncG4''[
        KineticXEScalar[]]
      ClockProfile^(0,1)[CTau[], CRho[]]^2
      (ClockProfile^(0,1)[CTau[], CRho[]]
            MetricA^(0,1)[CTau[], CRho[]] -
          (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile^(0,2)[CTau[], CRho[]] -
                ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]))
      (-4 + 4 ArealRadius^(0,1)[CTau[], CRho[]]^2 + 4 ArealRadius[CTau[], CRho[]]
            ArealRadius^(0,1)[CTau[], CRho[]]
            MetricA^(0,1)[CTau[], CRho[]] +
          8 ArealRadius[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] +
          4 ArealRadius^(1,0)[CTau[], CRho[]]^2 -
          4 ArealRadius[CTau[], CRho[]]
            ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
          ArealRadius[CTau[], CRho[]]^2 MetricA^(1,0)[CTau[], CRho[]]^2 +
          8 ArealRadius[CTau[], CRho[]]
            ArealRadius^(2,0)[CTau[], CRho[]] +
          4 MetricA[CTau[], CRho[]]^2 (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2 +
                2 ArealRadius[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]]) -
          2 ArealRadius[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]] -
          2 MetricA[CTau[], CRho[]] (2 ArealRadius^(0,1)[CTau[], CRho[]]^2 +
                4 (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2) + ArealRadius[CTau[], CRho[]]
                  (4 ArealRadius^(0,2)[CTau[], CRho[]] - 2 ArealRadius^(1,0)[CTau[], CRho[]]
                        MetricA^(1,0)[CTau[], CRho[]] + 8 ArealRadius^(2,0)[CTau[], CRho[]]) -
                ArealRadius[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]]))) /
  (ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^5) +
(FuncG4'[
        KineticXEScalar[]]
      ClockProfile^(2,0)[
        CTau[],
        CRho[]]
      (-4 + 4 ArealRadius^(0,1)[CTau[], CRho[]]^2 +
          4 ArealRadius[CTau[], CRho[]]
            ArealRadius^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]] +
          8 ArealRadius[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] +
          4 ArealRadius^(1,0)[CTau[], CRho[]]^2 -
          4 ArealRadius[CTau[], CRho[]]
            ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
          ArealRadius[CTau[], CRho[]]^2 MetricA^(1,0)[CTau[], CRho[]]^2 +
          8 ArealRadius[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]] +
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            4 MetricA[CTau[], CRho[]]^2
              (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2 +
                  2 ArealRadius[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]]) -
            2 ArealRadius[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]] -
            2 MetricA[CTau[], CRho[]] (2 ArealRadius^(0,1)[CTau[], CRho[]]^2 +
                  4 (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2) + ArealRadius[CTau[], CRho[]]
                    (4 ArealRadius^(0,2)[CTau[], CRho[]] - 2 ArealRadius^(1,0)[CTau[], CRho[]]
                          MetricA^(1,0)[CTau[], CRho[]] + 8 ArealRadius^(2,0)[CTau[], CRho[]]) -
                  ArealRadius[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]]))) /
  (ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^2) +
(2
      FuncG4''[
        KineticXEScalar[]]
      ClockProfile^(1,0)[CTau[], CRho[]]^2
      ClockProfile^(2,0)[
        CTau[],
        CRho[]]
      (-4 + 4 ArealRadius^(0,1)[CTau[], CRho[]]^2 +
            4 ArealRadius[CTau[], CRho[]]
              ArealRadius^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]] +
            8 ArealRadius[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] +
            4 ArealRadius^(1,0)[CTau[], CRho[]]^2 -
            4 ArealRadius[CTau[], CRho[]]
              ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
            ArealRadius[CTau[], CRho[]]^2 MetricA^(1,0)[CTau[], CRho[]]^2 +
            8 ArealRadius[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]] +
            4 MetricA[CTau[], CRho[]]^2 (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2 +
                  2 ArealRadius[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]]) -
            2 ArealRadius[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]] -
            2 MetricA[CTau[], CRho[]] (2 ArealRadius^(0,1)[CTau[], CRho[]]^2 +
                  4 (-1 + ArealRadius^(1,0)[CTau[], CRho[]]^2) + ArealRadius[CTau[], CRho[]]
                    (4 ArealRadius^(0,2)[CTau[], CRho[]] - 2 ArealRadius^(1,0)[CTau[], CRho[]]
                          MetricA^(1,0)[CTau[], CRho[]] + 8 ArealRadius^(2,0)[CTau[], CRho[]]) -
                  ArealRadius[CTau[], CRho[]]^2 MetricA^(2,0)[CTau[], CRho[]]))) /
  (ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^2) +
(2
      FuncG4'[
        KineticXEScalar[]]
      (ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]] -
            (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(1,0)[CTau[], CRho[]]
              ClockProfile^(1,0)[CTau[], CRho[]])
      (4 - 4 ArealRadius^(0,1)[CTau[], CRho[]]^2 - 4 ArealRadius[CTau[], CRho[]]
              ArealRadius^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]] -
            8 ArealRadius[CTau[], CRho[]] ArealRadius^(0,2)[CTau[], CRho[]] -
            4 ArealRadius^(1,0)[CTau[], CRho[]]^2 +
            4 ArealRadius[CTau[], CRho[]]
              ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] +
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ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
8 ArealRadius[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
4 MetricA[CTau[], CRho[]]^2 (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 +
2 ArealRadius[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
2 ArealRadius[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
2 MetricA[CTau[], CRho[]] (2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 +
4 (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2) + ArealRadius[CTau[], CRho[]]
(4 ArealRadius$^{(0,2)}$[CTau[], CRho[]] - 2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 8 ArealRadius$^{(2,0)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]]^3 (-1 + MetricA[CTau[], CRho[]])^3) -
(2
FuncG4′′[
KineticXEScalar[]]
ClockProfile$^{(0,1)}$[
CTau[],
CRho[]]
ClockProfile$^{(1,0)}$[
CTau[],
CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(1,1)}$[CTau[], CRho[]])
(4 - 4 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 - 4 ArealRadius[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
8 ArealRadius[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
4 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 +
4 ArealRadius[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
8 ArealRadius[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] -
4 MetricA[CTau[], CRho[]]^2 (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 +
2 ArealRadius[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
2 ArealRadius[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
2 MetricA[CTau[], CRho[]] (2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 +
4 (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2) + ArealRadius[CTau[], CRho[]]
(4 ArealRadius$^{(0,2)}$[CTau[], CRho[]] - 2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] + 8 ArealRadius$^{(2,0)}$[CTau[], CRho[]]) -
ArealRadius[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]]^2 (-1 + MetricA[CTau[], CRho[]])^4) +
(8
FuncG4′′[
KineticXEScalar[]]
ClockProfile$^{(1,0)}$[
CTau[],
CRho[]]
(-ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] +

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        (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(1,0)[CTau[], CRho[]]
          ClockProfile^(1,0)[CTau[], CRho[]])
      (-ClockProfile^(0,1)[CTau[], CRho[]] (2 MetricA^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[],
                CRho[]] - (-1 + MetricA[CTau[], CRho[]]) MetricA^(1,1)[CTau[], CRho[]]) +
        (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile^(0,2)[CTau[], CRho[]]
              MetricA^(1,0)[CTau[], CRho[]] + MetricA^(0,1)[CTau[], CRho[]]
              ClockProfile^(1,1)[CTau[], CRho[]] + 2 ClockProfile^(1,2)[CTau[], CRho[]] -
            2 MetricA[CTau[], CRho[]] ClockProfile^(1,2)[CTau[], CRho[]] -
            MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]] +
            MetricA[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[],
                CRho[]] - ClockProfile^(1,0)[CTau[], CRho[]] (MetricA^(1,0)[CTau[], CRho[]]^2 -
                  (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,0)[CTau[], CRho[]])))) /
    (ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^4) +
  (2
      FuncG4''[
        KineticXEScalar[]]
      ClockProfile^(0,1)[
        CTau[],
        CRho[]]
      (-ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] +
          2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(1,1)[CTau[], CRho[]])
      (-ClockProfile^(0,1)[CTau[], CRho[]]
            (2 MetricA^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
              (-1 + MetricA[CTau[], CRho[]]) MetricA^(1,1)[CTau[], CRho[]]) +
          (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile^(0,2)[CTau[], CRho[]]
              MetricA^(1,0)[CTau[], CRho[]] + MetricA^(0,1)[CTau[], CRho[]]
              ClockProfile^(1,1)[CTau[], CRho[]] + 2 ClockProfile^(1,2)[CTau[], CRho[]] -
            2 MetricA[CTau[], CRho[]] ClockProfile^(1,2)[CTau[], CRho[]] -
            MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]] +
            MetricA[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[],
                CRho[]] - ClockProfile^(1,0)[CTau[], CRho[]] (MetricA^(1,0)[CTau[], CRho[]]^2 -
                  (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,0)[CTau[], CRho[]])))) /
    (-1 + MetricA[CTau[], CRho[]])^5 - (4 FuncG4''[KineticXEScalar[]]
      ClockProfile^(1,0)[
        CTau[],
        CRho[]]
      ClockProfile^(2,0)[
        CTau[],
        CRho[]]
      (-ClockProfile^(0,1)[CTau[], CRho[]]
            (2 MetricA^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
              (-1 + MetricA[CTau[], CRho[]]) MetricA^(1,1)[CTau[], CRho[]]) +
          (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile^(0,2)[CTau[], CRho[]]
              MetricA^(1,0)[CTau[], CRho[]] + MetricA^(0,1)[CTau[], CRho[]]
              ClockProfile^(1,1)[CTau[], CRho[]] + 2 ClockProfile^(1,2)[CTau[], CRho[]] -
            2 MetricA[CTau[], CRho[]] ClockProfile^(1,2)[CTau[], CRho[]] -
            MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]] +
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MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[],
CRho[]] - ClockProfile$^{(1,0)}$[CTau[], CRho[]] (MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]])))) /
(-1 + MetricA[CTau[], CRho[]])^3 - (4 FuncG4′[KineticXEScalar[]]
(-((2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
((ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(0,1)}$[CTau[], CRho[]]) /
(-1 + MetricA[CTau[], CRho[]]) + $\frac{1}{2}$ ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(-2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,0)}$[CTau[], CRho[]] +
(ArealRadius[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(0,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[],
CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[],
CRho[]]))) / (-1 + MetricA[CTau[], CRho[]])^2) +
ArealRadius[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
((ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) /
(2 - 2 MetricA[CTau[], CRho[]]) + ClockProfile$^{(1,1)}$[
CTau[], CRho[]]))) / ArealRadius[CTau[], CRho[]]) -
(ArealRadius[CTau[], CRho[]] ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 MetricA$^{(0,1)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) (MetricA$^{(0,2)}$[CTau[], CRho[]] -
MetricA$^{(1,0)}$[CTau[], CRho[]]^2)) + (-1 + MetricA[CTau[], CRho[]])
(MetricA$^{(0,1)}$[CTau[], CRho[]] (-3 ClockProfile$^{(0,2)}$[CTau[], CRho[]] +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,3)}$[CTau[], CRho[]] -
3 MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]])))) /
(2 (-1 + MetricA[CTau[], CRho[]])^3) + (ArealRadius[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(1,1)}$[CTau[], CRho[]]) - (-1 + MetricA[CTau[], CRho[]])
(2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] + 2
ClockProfile$^{(1,2)}$[CTau[], CRho[]] - 2 MetricA[CTau[], CRho[]]
ClockProfile$^{(1,2)}$[CTau[], CRho[]] - MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] + MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] (MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]])))) /
(2 (-1 + MetricA[CTau[], CRho[]])^2))) / (ArealRadius[CTau[], CRho[]]^2
(1 - MetricA[CTau[], CRho[]])) - (2
FuncG4′[
KineticXEScalar[]]
(-4 ArealRadius$^{(0,1)}$[CTau[], CRho[]]
((ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 ClockProfile$^{(0,1)}$[CTau[], CRho[]]) /
(-1 + MetricA[CTau[], CRho[]]) +

$\frac{1}{2}$ ArealRadius$^{(0,1)}$[CTau[], CRho[]] (-2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]] + (ArealRadius[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[],
CRho[]] - ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[
CTau[], CRho[]]))) / (-1 + MetricA[CTau[], CRho[]])^{2}) +
ArealRadius[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
((ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) /
(2 - 2 MetricA[CTau[], CRho[]]) + ClockProfile$^{(1,1)}$[CTau[], CRho[]])) -
(ArealRadius[CTau[], CRho[]]^{2} ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 MetricA$^{(0,1)}$[CTau[], CRho[]]^{2} -
(-1 + MetricA[CTau[], CRho[]]) (MetricA$^{(0,2)}$[CTau[], CRho[]] -
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2})) + (-1 + MetricA[CTau[], CRho[]])
(MetricA$^{(0,1)}$[CTau[], CRho[]] (-3 ClockProfile$^{(0,2)}$[CTau[], CRho[]] +
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,3)}$[CTau[], CRho[]] -
3 MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]])))) /
(-1 + MetricA[CTau[], CRho[]])^{3} + (ArealRadius[CTau[], CRho[]]^{2}
ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(1,1)}$[CTau[], CRho[]]) - (-1 + MetricA[CTau[], CRho[]])
(2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] + 2
ClockProfile$^{(1,2)}$[CTau[], CRho[]] - 2 MetricA[CTau[], CRho[]]
ClockProfile$^{(1,2)}$[CTau[], CRho[]] - MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] + MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] (MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]])))) /
(-1 + MetricA[CTau[], CRho[]])^{2})) / (ArealRadius[CTau[], CRho[]]^{3}
(1 - MetricA[CTau[], CRho[]])) +
(FuncG4′′[KineticXEScalar[]] ClockProfile$^{(1,0)}$[
CTau[],
CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(1,1)}$[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
2 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) -
4 (-1 + MetricA[CTau[], CRho[]])
(MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,1)}$[CTau[], CRho[]]))) /
(-1 + MetricA[CTau[], CRho[]])^{4} + (8 FuncG4′′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[
CTau[]

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          CTau[],
          CRho[]]
        (ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]] -
            (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(1,0)[CTau[], CRho[]]
              ClockProfile^(1,0)[CTau[], CRho[]])
        (ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]^2 -
            2 (-1 + MetricA[CTau[], CRho[]])
              (MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
                  (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(2,1)[CTau[], CRho[]]))) /
    (ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^4) +
  (2
        FuncG4''[
          KineticXEScalar[]]
        ClockProfile^(0,1)[
          CTau[],
          CRho[]]
        (ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(0,1)[CTau[], CRho[]] -
            (-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile^(0,2)[CTau[], CRho[]] -
                  ClockProfile^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]))
        (ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]^2 -
            2 (-1 + MetricA[CTau[], CRho[]])
              (MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
                  (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(2,1)[CTau[], CRho[]]))) /
    (-1 + MetricA[CTau[], CRho[]])^5 -
  (2
        FuncG4''[
          KineticXEScalar[]]
        ClockProfile^(1,0)[
          CTau[],
          CRho[]]
        (ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
            2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(1,1)[CTau[], CRho[]])
        (ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]^2 -
            2 (-1 + MetricA[CTau[], CRho[]])
              (MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
                  (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(2,1)[CTau[], CRho[]]))) /
    (-1 + MetricA[CTau[], CRho[]])^4 +
  (4
        FuncG4''[
          KineticXEScalar[]]
        ClockProfile^(0,1)[
          CTau[],
          CRho[]]
        (ArealRadius^(0,1)[CTau[], CRho[]] ClockProfile^(0,1)[CTau[], CRho[]] -
            (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(1,0)[CTau[], CRho[]]
              ClockProfile^(1,0)[CTau[], CRho[]])
        (ClockProfile^(0,1)[CTau[], CRho[]] (-3 MetricA^(1,0)[CTau[], CRho[]]^2 +
                  2 (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,0)[CTau[], CRho[]]) +
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4 (-1 + MetricA[CTau[], CRho[]])
(MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,1)}$[CTau[], CRho[]]))) /
(ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])^4) +
(FuncG4′′[
KineticXEScalar[]]
ClockProfile$^{(0,1)}$[
CTau[],
CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]))
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (-3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 +
2 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) +
4 (-1 + MetricA[CTau[], CRho[]])
(MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,1)}$[CTau[], CRho[]]))) /
(-1 + MetricA[CTau[], CRho[]])^5 - (2 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(0,1)}$[
CTau[],
CRho[]]
(4 (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^3 -
2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] (2 + ArealRadius[CTau[], CRho[]]^2
(2 MetricA$^{(0,1)}$[CTau[], CRho[]]^2 + MetricA$^{(0,2)}$[CTau[], CRho[]]) -
2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 - MetricA[CTau[], CRho[]]
(6 + ArealRadius[CTau[], CRho[]]^2 MetricA$^{(0,2)}$[CTau[], CRho[]] -
6 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 + 2 ArealRadius[CTau[], CRho[]]
(ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
3 ArealRadius$^{(2,0)}$[CTau[], CRho[]])) + ArealRadius[CTau[], CRho[]]
(ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]) + MetricA[CTau[], CRho[]]^2
(6 - 6 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 + ArealRadius[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
6 ArealRadius[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]]) +
2 MetricA[CTau[], CRho[]]^3 (-1 + ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 +
ArealRadius[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]])) +
ArealRadius[CTau[], CRho[]] (4 (-1 + MetricA[CTau[], CRho[]])^3
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
(MetricA$^{(0,1)}$[CTau[], CRho[]] (3 ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) (-2 ArealRadius$^{(0,3)}$[CTau[], CRho[]] +
MetricA$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
2 ArealRadius$^{(2,1)}$[CTau[], CRho[]] +
2 MetricA[CTau[], CRho[]] ArealRadius$^{(2,1)}$[CTau[], CRho[]])) +
ArealRadius[CTau[], CRho[]]^2 (MetricA$^{(0,1)}$[CTau[], CRho[]]

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                 ArealRadius[CTau[], CRho[]]  (MetricA      [CTau[], CRho[]]
                         (MetricA^(1,0)[CTau[], CRho[]]^2 - (-1 + MetricA[CTau[], CRho[]])
                               MetricA^(2,0)[CTau[], CRho[]]) - (-1 + MetricA[CTau[], CRho[]])
                         (MetricA^(1,0)[CTau[], CRho[]] MetricA^(1,1)[CTau[], CRho[]] -
                             (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,1)[CTau[], CRho[]]))))) /
  (ArealRadius[CTau[], CRho[]]^3 (-1 + MetricA[CTau[], CRho[]])^4) -
(4
    FuncG4′[
      KineticXEScalar[]]
    (-((2 ArealRadius^(1,0)[CTau[], CRho[]] ((ArealRadius^(0,1)[CTau[], CRho[]]
                               (ClockProfile^(0,1)[CTau[], CRho[]] (2 (-1 + MetricA[CTau[], CRho[]])
                                           ArealRadius^(1,0)[CTau[], CRho[]] + ArealRadius[
                                             CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]) -
                                   2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
                                     ClockProfile^(1,1)[CTau[], CRho[]])) /
                           (2 (-1 + MetricA[CTau[], CRho[]])^2) + ArealRadius^(1,0)[CTau[], CRho[]]
                           (-ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,0)[CTau[], CRho[]] +
                               ArealRadius[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]]))) /
              ArealRadius[CTau[], CRho[]]) - (ArealRadius[CTau[], CRho[]]
              ArealRadius^(0,1)[CTau[], CRho[]] ((MetricA^(1,0)[CTau[], CRho[]]
                           (ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] -
                               2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(1,1)[CTau[], CRho[]])) /
                       (2 (-1 + MetricA[CTau[], CRho[]])^2) + ClockProfile^(2,1)[CTau[], CRho[]])) /
          (-1 + MetricA[CTau[], CRho[]]) + ArealRadius[CTau[], CRho[]]
          ArealRadius^(1,0)[CTau[], CRho[]]
          ClockProfile^(3,0)[CTau[], CRho[]])) /
  ArealRadius[CTau[], CRho[]]^2 + (4
    FuncG4′[
      KineticXEScalar[]]
    ((2 ArealRadius^(1,0)[CTau[], CRho[]]^2 (-ArealRadius^(0,1)[CTau[], CRho[]]
                       ClockProfile^(0,1)[CTau[], CRho[]] + (-1 + MetricA[CTau[], CRho[]])
                       ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,0)[CTau[], CRho[]])) /
          (ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])) -
        (ArealRadius^(1,0)[CTau[], CRho[]] (ArealRadius^(0,1)[CTau[], CRho[]]
                       (3 ClockProfile^(0,1)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] - 4
                             (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(1,1)[CTau[], CRho[]]) +
                     2 (-1 + MetricA[CTau[], CRho[]]) (-ClockProfile^(0,1)[CTau[], CRho[]]
                             ArealRadius^(1,1)[CTau[], CRho[]] + (-1 + MetricA[CTau[], CRho[]])
                             (ClockProfile^(1,0)[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]] +
                                 2 ArealRadius^(1,0)[CTau[], CRho[]]
                                   ClockProfile^(2,0)[CTau[], CRho[]])))) /
          (2 (-1 + MetricA[CTau[], CRho[]])^2) - (ArealRadius[CTau[], CRho[]]
              ArealRadius^(0,1)[CTau[], CRho[]] (ClockProfile^(0,1)[CTau[], CRho[]]
                     (3 MetricA^(1,0)[CTau[], CRho[]]^2 - 2 (-1 + MetricA[CTau[], CRho[]])
                           MetricA^(2,0)[CTau[], CRho[]]) - 4 (-1 + MetricA[CTau[], CRho[]])
                     (MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
                         (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(2,1)[CTau[], CRho[]]))) /
          (4 (-1 + MetricA[CTau[], CRho[]])^3) + ArealRadius[CTau[], CRho[]]
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        (4 (-1 + MetricA[CTau[], CRho[]])^3) + ArealRadius[CTau[], CRho[]]
        ArealRadius^(1,0)[CTau[], CRho[]]
        ClockProfile^(3,0)[CTau[], CRho[]])) /
ArealRadius[CTau[], CRho[]]^2 + (8
  FuncG4′[
   KineticXEScalar[]]
  (ArealRadius^(2,0)[CTau[], CRho[]]
        (-ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,0)[CTau[], CRho[]] +
          ArealRadius[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]]) -
      (2 ArealRadius^(1,0)[CTau[], CRho[]] ((ArealRadius^(0,1)[CTau[], CRho[]]
                (ClockProfile^(0,1)[CTau[], CRho[]] (2 (-1 + MetricA[CTau[], CRho[]])
                      ArealRadius^(1,0)[CTau[], CRho[]] +
                     ArealRadius[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]) -
                  2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
                   ClockProfile^(1,1)[CTau[], CRho[]])) /
              (2 (-1 + MetricA[CTau[], CRho[]])^2) + ArealRadius^(1,0)[CTau[], CRho[]]
              (-ArealRadius^(1,0)[CTau[], CRho[]] ClockProfile^(1,0)[CTau[], CRho[]] +
                ArealRadius[CTau[], CRho[]] ClockProfile^(2,0)[CTau[], CRho[]]))) /
        ArealRadius[CTau[], CRho[]] + ((-1 + MetricA[CTau[], CRho[]])
            ArealRadius^(1,1)[CTau[], CRho[]] (ClockProfile^(0,1)[CTau[], CRho[]]
               (2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(1,0)[CTau[], CRho[]] +
                 ArealRadius[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]) -
              2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
               ClockProfile^(1,1)[CTau[], CRho[]]) + ArealRadius^(0,1)[CTau[], CRho[]]
            (ClockProfile^(0,1)[CTau[], CRho[]] (-((-1 + MetricA[CTau[], CRho[]])
                    ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]]) +
                 2 (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius^(2,0)[CTau[], CRho[]] +
                 ArealRadius[CTau[], CRho[]] (-2 MetricA^(1,0)[CTau[], CRho[]]^2 +
                    (-1 + MetricA[CTau[], CRho[]]) MetricA^(2,0)[CTau[], CRho[]])) +
              ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
               (3 MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
                 2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(2,1)[CTau[], CRho[]]))) /
        (2 (-1 + MetricA[CTau[], CRho[]])^3) + ArealRadius^(1,0)[CTau[], CRho[]]
        (-ClockProfile^(1,0)[CTau[], CRho[]] ArealRadius^(2,0)[CTau[], CRho[]] +
          ArealRadius[CTau[], CRho[]] ClockProfile^(3,0)[CTau[], CRho[]]))) /
ArealRadius[CTau[], CRho[]]^2 - (4 FuncG4′[KineticXEScalar[]]
  (-((ArealRadius^(0,1)[CTau[], CRho[]] (ClockProfile^(0,1)[CTau[], CRho[]]
                (4 (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius^(1,0)[CTau[], CRho[]]^2 +
                  2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
                   ArealRadius^(1,0)[CTau[], CRho[]] MetricA^(1,0)[CTau[], CRho[]] +
                  ArealRadius[CTau[], CRho[]]^2 MetricA^(1,0)[CTau[], CRho[]]^2) -
               2 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
                (2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius^(1,0)[CTau[], CRho[]]
                   ClockProfile^(1,1)[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]
                   (MetricA^(1,0)[CTau[], CRho[]] ClockProfile^(1,1)[CTau[], CRho[]] -
                     (-1 + MetricA[CTau[], CRho[]]) ClockProfile^(2,1)[CTau[],
                       CRho[]])))) / (2 (-1 + MetricA[CTau[], CRho[]])^3)) +
```

ArealRadius$^{(1,0)}$[CTau[], CRho[]] (2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(1,0)}$[CTau[], CRho[]] - 2 ArealRadius[CTau[], CRho[]]
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]]^2 ClockProfile$^{(3,0)}$[CTau[], CRho[]]))) /
ArealRadius[CTau[], CRho[]]^3 + (FuncG4′[KineticXEScalar[]]
(2 ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 MetricA$^{(0,1)}$[CTau[], CRho[]]
(3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 - (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(2,0)}$[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
(-4 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,1)}$[CTau[], CRho[]])) +
(-1 + MetricA[CTau[], CRho[]]) (-8 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
4 ClockProfile$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
4 MetricA[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] - 8 MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,2)}$[CTau[], CRho[]] + 8 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,2)}$[CTau[], CRho[]] +
3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
3 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(2,0)}$[CTau[], CRho[]] + 2 ClockProfile$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] - 4 MetricA[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
2 MetricA[CTau[], CRho[]]^2 ClockProfile$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 2 ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] (MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) -
4 ClockProfile$^{(0,2)}$[CTau[], CRho[]] (2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) -
2 MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(2,1)}$[CTau[], CRho[]] +
2 MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(2,1)}$[CTau[], CRho[]] - 4 ClockProfile$^{(2,2)}$[CTau[], CRho[]] +
8 MetricA[CTau[], CRho[]] ClockProfile$^{(2,2)}$[CTau[], CRho[]] -
4 MetricA[CTau[], CRho[]]^2 ClockProfile$^{(2,2)}$[CTau[], CRho[]] +
2 MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(3,0)}$[CTau[], CRho[]] -
4 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(3,0)}$[CTau[], CRho[]] + 2 MetricA[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(3,0)}$[CTau[], CRho[]]))) /
(-1 + MetricA[CTau[], CRho[]])^4 + (FuncG4′[KineticXEScalar[]]
(-((ArealRadius$^{(0,1)}$[CTau[], CRho[]] (ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(8 (-1 + MetricA[CTau[], CRho[]])^2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2 +
6 ArealRadius[CTau[], CRho[]] (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]]^2 (3 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
2 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[],
CRho[]])) - 4 ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]]) (2 (-1 + MetricA[CTau[], CRho[]])

(-1 + MetricA[CTau[], CRho[]]) (2 (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[],
CRho[]] + ArealRadius[CTau[], CRho[]] (MetricA$^{(1,0)}$[CTau[],
CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] - (-1 + MetricA[
CTau[], CRho[]]) ClockProfile$^{(2,1)}$[CTau[], CRho[]])))) /
(-1 + MetricA[CTau[], CRho[]])^{3}) + 4 ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(2 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(1,0)}$[CTau[], CRho[]] -
2 ArealRadius[CTau[], CRho[]] ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] ((ClockProfile$^{(0,1)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]]) / (-1 + MetricA[CTau[], CRho[]]) -
ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]] ClockProfile$^{(3,0)}$[CTau[], CRho[]])))) /
ArealRadius[CTau[], CRho[]]^{3} + (FuncG4′[KineticXEScalar[]]
(-ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(MetricA$^{(0,1)}$[CTau[], CRho[]]
(23 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} - 6 (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(2,0)}$[CTau[], CRho[]]) + 4 (-1 + MetricA[CTau[], CRho[]])
(-4 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,1)}$[CTau[], CRho[]])) +
(-1 + MetricA[CTau[], CRho[]]) (2 ClockProfile$^{(0,2)}$[CTau[], CRho[]]
(7 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} - 2 (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(2,0)}$[CTau[], CRho[]]) + ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] (-3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} +
2 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) +
4 MetricA$^{(0,1)}$[CTau[], CRho[]] (4 MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(2,1)}$[CTau[], CRho[]]) - 4 (-1 + MetricA[CTau[], CRho[]])
(2 ClockProfile$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,2)}$[CTau[], CRho[]] +
MetricA$^{(1,0)}$[CTau[], CRho[]] (4 ClockProfile$^{(1,2)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(3,0)}$[CTau[], CRho[]]))))) /
(2 (1 - MetricA[CTau[], CRho[]]) (-1 + MetricA[CTau[], CRho[]])^{3}) -
(FuncG4′[
KineticXEScalar[]]
(3 ClockProfile$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]
(3 MetricA$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) MetricA$^{(1,1)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) (-6 ClockProfile$^{(0,2)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} + ClockProfile$^{(1,0)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]]^{3} - 2 MetricA$^{(0,1)}$[CTau[], CRho[]]
(4 MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,1)}$[CTau[], CRho[]]) +
2 (-1 + MetricA[CTau[], CRho[]])
(2 ClockProfile$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] -
MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} ClockProfile$^{(2,0)}$[CTau[], CRho[]] -

MetricA [CTau[], CRho[]] ClockProfile [CTau[], CRho[]] -
2 (-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(2,2)}$[CTau[], CRho[]] +
MetricA$^{(1,0)}$[CTau[], CRho[]] (4 ClockProfile$^{(1,2)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ClockProfile$^{(3,0)}$[CTau[], CRho[]]))))) /
(-1 + MetricA[CTau[], CRho[]])^{4} - (8 FuncG4′[KineticXEScalar[]]
((2 (-1 + MetricA[CTau[], CRho[]])^{2} ArealRadius$^{(1,0)}$[CTau[], CRho[]]^{2}
(-ArealRadius$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(0,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,0)}$[CTau[], CRho[]])) / ArealRadius[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) (ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] - ClockProfile$^{(0,1)}$[CTau[], CRho[]]
(2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(2,0)}$[CTau[], CRho[]])) -
(-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(-2 ClockProfile$^{(0,1)}$[CTau[], CRho[]] ArealRadius$^{(1,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]])
(3 ClockProfile$^{(1,0)}$[CTau[], CRho[]] ArealRadius$^{(2,0)}$[CTau[], CRho[]] +
2 ArealRadius$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]))) -
ArealRadius[CTau[], CRho[]] (ArealRadius$^{(0,1)}$[CTau[], CRho[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) +
(-1 + MetricA[CTau[], CRho[]]) (-2 MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(1,1)}$[CTau[], CRho[]] + (-1 + MetricA[CTau[], CRho[]])
ClockProfile$^{(2,1)}$[CTau[], CRho[]])) - (-1 + MetricA[CTau[], CRho[]])
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 MetricA$^{(1,0)}$[CTau[], CRho[]]
ArealRadius$^{(1,1)}$[CTau[], CRho[]] - (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(2,1)}$[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
(-2 ArealRadius$^{(1,1)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) (2 ArealRadius$^{(2,0)}$[CTau[], CRho[]]
ClockProfile$^{(2,0)}$[CTau[], CRho[]] + ClockProfile$^{(1,0)}$[CTau[],
CRho[]] ArealRadius$^{(3,0)}$[CTau[], CRho[]] + ArealRadius$^{(1,0)}$[
CTau[], CRho[]] ClockProfile$^{(3,0)}$[CTau[], CRho[]])))))) /
(ArealRadius[CTau[], CRho[]]^{2} (-1 + MetricA[CTau[], CRho[]])^{3}) -
(2
FuncG4′[
KineticXEScalar[]]
(ClockProfile$^{(0,1)}$[CTau[], CRho[]] (2 MetricA$^{(0,1)}$[CTau[], CRho[]]
(3 MetricA$^{(1,0)}$[CTau[], CRho[]]^{2} - (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(2,0)}$[CTau[], CRho[]]) + (-1 + MetricA[CTau[], CRho[]])
(-4 MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,1)}$[CTau[], CRho[]])) +
(-1 + MetricA[CTau[], CRho[]]) (-4 MetricA$^{(0,1)}$[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]] -
2 ClockProfile$^{(1,1)}$[CTau[], CRho[]] MetricA$^{(1,1)}$[CTau[], CRho[]] +
2 MetricA[CTau[], CRho[]] ClockProfile$^{(1,1)}$[CTau[], CRho[]]
MetricA$^{(1,1)}$[CTau[], CRho[]] - 4 MetricA$^{(1,0)}$[CTau[], CRho[]]

ClockProfile$^{(1,2)}$[CTau[], CRho[]] + 4 MetricA[CTau[], CRho[]]
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(1,2)}$[CTau[], CRho[]] +
2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 ClockProfile$^{(2,0)}$[CTau[], CRho[]] -
2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]]^2
ClockProfile$^{(2,0)}$[CTau[], CRho[]] +
2 ClockProfile$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
4 MetricA[CTau[], CRho[]] ClockProfile$^{(2,0)}$[CTau[], CRho[]]
MetricA$^{(2,0)}$[CTau[], CRho[]] + 2 MetricA[CTau[], CRho[]]^2
ClockProfile$^{(2,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] -
2 ClockProfile$^{(0,2)}$[CTau[], CRho[]] (2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
(-1 + MetricA[CTau[], CRho[]]) MetricA$^{(2,0)}$[CTau[], CRho[]]) -
MetricA$^{(0,1)}$[CTau[], CRho[]] ClockProfile$^{(2,1)}$[CTau[], CRho[]] +
MetricA[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]]
ClockProfile$^{(2,1)}$[CTau[], CRho[]] - 2 ClockProfile$^{(2,2)}$[CTau[], CRho[]] +
4 MetricA[CTau[], CRho[]] ClockProfile$^{(2,2)}$[CTau[], CRho[]] -
2 MetricA[CTau[], CRho[]]^2 ClockProfile$^{(2,2)}$[CTau[], CRho[]] +
MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(3,0)}$[CTau[], CRho[]] -
2 MetricA[CTau[], CRho[]] MetricA$^{(1,0)}$[CTau[], CRho[]] ClockProfile$^{(3,0)}$[
CTau[], CRho[]] + MetricA[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]
ClockProfile$^{(3,0)}$[CTau[], CRho[]] + ClockProfile$^{(1,0)}$[CTau[], CRho[]]
(2 MetricA$^{(1,0)}$[CTau[], CRho[]]^3 - 3 (-1 + MetricA[CTau[], CRho[]])
MetricA$^{(1,0)}$[CTau[], CRho[]] MetricA$^{(2,0)}$[CTau[], CRho[]] +
(-1 + MetricA[CTau[], CRho[]])^2 MetricA$^{(3,0)}$[CTau[], CRho[]])))) /
(-1 + MetricA[CTau[], CRho[]])^4 + (2 FuncG4′[KineticXEScalar[]]
ClockProfile$^{(1,0)}$[
CTau[],
CRho[]]
(-4 (-1 + MetricA[CTau[], CRho[]])^3 ArealRadius$^{(1,0)}$[CTau[], CRho[]]^3 -
2 ArealRadius[CTau[], CRho[]]
(-1 + MetricA[CTau[], CRho[]])^2
ArealRadius$^{(1,0)}$[CTau[], CRho[]]^2
MetricA$^{(1,0)}$[CTau[], CRho[]] +
2 (-1 + MetricA[CTau[], CRho[]]) ArealRadius$^{(1,0)}$[CTau[], CRho[]]
(2 + 2 MetricA[CTau[], CRho[]]^2 -
2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 - ArealRadius[CTau[], CRho[]]
ArealRadius$^{(0,1)}$[CTau[], CRho[]] MetricA$^{(0,1)}$[CTau[], CRho[]] -
2 ArealRadius[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] -
ArealRadius[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]]^2 -
ArealRadius[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]] +
MetricA[CTau[], CRho[]] (-4 + 2 ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 +
2 ArealRadius[CTau[], CRho[]] ArealRadius$^{(0,2)}$[CTau[], CRho[]] +
ArealRadius[CTau[], CRho[]]^2 MetricA$^{(2,0)}$[CTau[], CRho[]])) +
ArealRadius[CTau[], CRho[]] (2 (-1 + MetricA[CTau[], CRho[]])
ArealRadius$^{(0,1)}$[CTau[], CRho[]]^2 MetricA$^{(1,0)}$[CTau[], CRho[]] -
2 ArealRadius$^{(0,1)}$[CTau[], CRho[]] (2 (-1 + MetricA[CTau[], CRho[]])^2
ArealRadius$^{(1,1)}$[CTau[], CRho[]] + ArealRadius[CTau[], CRho[]]

$(2\ \text{MetricA}^{(0,1)}[\text{CTau}[],\ \text{CRho}[]]\ \text{MetricA}^{(1,0)}[\text{CTau}[],\ \text{CRho}[]] -$
$(-1 + \text{MetricA}[\text{CTau}[],\ \text{CRho}[]])\ \text{MetricA}^{(1,1)}[\text{CTau}[],\ \text{CRho}[]])) +$
$\text{ArealRadius}[\text{CTau}[],\ \text{CRho}[]]\ (4\ (-1 + \text{MetricA}[\text{CTau}[],\ \text{CRho}[]])$
$\text{ArealRadius}^{(0,2)}[\text{CTau}[],\ \text{CRho}[]]\ \text{MetricA}^{(1,0)}[\text{CTau}[],\ \text{CRho}[]] +$
$2\ (-1 + \text{MetricA}[\text{CTau}[],\ \text{CRho}[]])\ (\text{MetricA}^{(0,1)}[\text{CTau}[],\ \text{CRho}[]]$
$\text{ArealRadius}^{(1,1)}[\text{CTau}[],\ \text{CRho}[]] + (-1 + \text{MetricA}[\text{CTau}[],\ \text{CRho}[]])$
$(-2\ \text{ArealRadius}^{(1,2)}[\text{CTau}[],\ \text{CRho}[]] + \text{MetricA}^{(1,0)}[\text{CTau}[],\ \text{CRho}[]]$
$\text{ArealRadius}^{(2,0)}[\text{CTau}[],\ \text{CRho}[]] + 2\ (-1 + \text{MetricA}[\text{CTau}[],$
$\text{CRho}[]])\ \text{ArealRadius}^{(3,0)}[\text{CTau}[],\ \text{CRho}[]])) +$
$\text{ArealRadius}[\text{CTau}[],\ \text{CRho}[]]\ (\text{MetricA}^{(1,0)}[\text{CTau}[],\ \text{CRho}[]]^3 -$
$2\ (-1 + \text{MetricA}[\text{CTau}[],\ \text{CRho}[]])$
$\text{MetricA}^{(1,0)}[\text{CTau}[],\ \text{CRho}[]]\ \text{MetricA}^{(2,0)}[\text{CTau}[],\ \text{CRho}[]] +$
$(-1 + \text{MetricA}[\text{CTau}[],\ \text{CRho}[]])^2\ \text{MetricA}^{(3,0)}[\text{CTau}[],\ \text{CRho}[]]))))\ /$
$(\text{ArealRadius}[\text{CTau}[],\ \text{CRho}[]]^3\ (-1 + \text{MetricA}[\text{CTau}[],\ \text{CRho}[]])^3) ==$
0

KineticXEScalar

```
SphericalEquations["KineticXEScalar"]
```

$$\frac{\text{ClockProfile}^{(0,1)}[\text{CTau}[],\text{CRho}[]]^2}{1-\text{MetricA}[\text{CTau}[],\text{CRho}[]]} + \text{ClockProfile}^{(1,0)}[\text{CTau}[],\ \text{CRho}[]]^2$$